TrayGuard

A Local Tray Training And Assessment System

Final Design Review

EC ENGR 180DW – Systems Design

University of California, Los Angeles

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Note: Each section of this report lists in parentheses the team member(s) who authored that section, following the FDR report outline convention.

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1 Executive Summary

(Matthew)

Tray-content errors during reconstruction of reusable stainless steel surgical instrument trays are a recurring patient-safety risk in sterile processing departments (SPDs). Root-cause analysis identified six contributing categories—training gaps, incomplete count sheets, local tray variation, nomenclature inconsistency, production pressure, and weak feedback loops—all driven by the structural absence of professional licensure and competency infrastructure in the SPD workforce.

Certification proves a starting point rather than local readiness. National certification tests assess general instrument knowledge but do not cover the local names, aliases, lookalike pairs, and tray-specific counts that vary by hospital and surgeon. Studies of certified sterile-processing professionals have found mean pre-test scores as low as 41% on novel local tasks, and certification failure rates of 28–50% confirm that the technical demands are substantive even before accounting for local tray variation. The target population for TrayGuard is therefore the technician who holds certification but remains unprepared for the local tray configurations they face daily.

The project initially pursued a computer-vision tray checker. Seven staged detection experiments (800–80 lumens, matte/reflective backgrounds, separated/overlay layouts) showed that YOLO11s detection degrades under every variation that SPD workflows present, with mAP50-95 falling to 0.389 under occlusion. Real-instrument experiments confirmed that position cues dominate over shape learning, making lighting-transfer conclusions unreliable without randomized layouts. Deployment-grade recognition would require site-specific training data, per-manufacturer validation, and workflow testing beyond the project's available resources.

The project pivoted to a training-centered platform: TrayGuard, a local tray training and assessment system using printable AprilTag instrument cards and a seven-step learning loop (intake, pre-test, study cards, quiz, practice sort, post-test, summary). A file-backed tray module defines instruments, aliases, required counts, distractors, lookalike pairs, and assessment variants. A scoring engine classifies each item into correct, missing, extra, wrong, misidentified, and wrong-count categories with confidence metadata.

A pre/post study with the seeded tray module would provide Kirkpatrick Level 2 evidence (simulated learning) with bounded claims. The first study uses a student convenience sample as a conservative baseline: if the measurement instrument cannot detect learning in novices, it will not work with certified technicians. The dependent variable shifts by population—absolute accuracy gain for the baseline, error-type reduction and confidence calibration for the technician target. Workplace transfer and clinical adoption remain future validation. The recommended next step is to build, test, and report the prototype study within the current scope.

2 Introduction And Project Overview

(Matthew)

Tray content errors during surgical instrument reconstruction are recurring operational failures that can delay surgery and compromise patient safety. The workforce faces structural pressure from growing demand, high turnover, and limited training capacity. Current training blends online coursework, classroom instruction, and hands-on externships that are capacity-constrained by clinical-site availability, clinical supervisor attention, and local teaching materials. Certification tests assess general instrument knowledge but do not cover the local names, aliases, lookalike pairs, and tray-specific counts that vary by hospital. Studies of certified sterile-processing professionals have found pre-test scores as low as 41% on novel local tasks [?], confirming that certification is a starting point rather than a guarantee of local readiness.

TrayGuard originally targeted a computer-vision-based tray checker: a camera above the assembly workstation would detect each instrument as the technician placed it, compare detections against the tray list, and return a go or no-go verdict before sealing. The CV approach failed early because it required the project to validate detection accuracy, lighting tolerance, occluded-item handling, and workflow integration inside a live SPD environment, resources not available within our project scope. The project pivoted to a training-centered alternative that preserves the same error categories (missing, wrong, extra, misidentified, and miscounted items) but measures them in a simulated local tray task using AprilTag instrument cards, retrieval study, and feedback modes.

This Functional Design Review serves as the literature-grounded design handoff for a future implementation team. It defines the problem, justifies the training-centered solution, specifies requirements and system architecture, provides a study plan for baseline evaluation (with a target-population path for SPD technicians), and documents the scope boundaries.

3 Problem Definition And Scope (Matthew, Andy, Qiyu)

3.1 Precise Problem Statement

Errors can occur throughout the sterile processing workflow: a technician may confuse two visually similar instruments, place the wrong size or pattern into a tray, omit or add an instrument, use the wrong count sheet, or fail to catch a documentation problem. In the OR, those mistakes appear as missing, wrong, extra, damaged, contaminated, or otherwise unavailable instruments.

Published rates vary by hospital, measurement method, and error definition, but the available evidence suggests that these are recurring operational failures, not isolated accidents. Alfred et al. analyzed 3,900 tray defects across 41,799 cases, or roughly 9 defects per 100 cases, and found that 55.0% of recorded defects occurred during assembly [?]. Zhu et al. found 398 packaging errors among 33,839 surgical instrument packages, or about 1.2% of packages. A detailed breakdown reveals nine categories: instrument in wrong specification (right type, wrong variant; 44.0%), instrument missing (19.4%), package incomplete (17.6%), instrument malfunction (6.8%), wrong packaging material (3.8%), box and cover mismatched (3.5%), wrong packaging tag (2.0%), indicator card missing (1.5%), and wrong count of instrument (1.5%) [?]. Nichol et al. observed 236 surgical instrument errors affecting 147 cases, with most errors tied to visualization work such as inspection, identification, function checking, and tray assembly. The resulting OR delays averaged 10.16 minutes per affected case, at an estimated annual cost of $6.75 million to $9.42 million [?]. Nichol and Saari further identified 368 patient safety notices containing 419 surgical instrument errors over 12 months, of which 83% involved failures in inspection, tracking, or identification [?].

The general problem is therefore:

> Sterile processing workflows do not always reliably produce complete, correct, functional, sterile, and ready-for-use surgical instrument sets.

3.2 Practice Context

The problem centers on reusable stainless steel surgical instruments, which make up the majority of SPD-handled inventory. Unlike single-use devices or textile goods, these instruments must be visually identified, sorted, and matched to tray lists after every use, and their metallic surfaces, subtle dimensional differences, and reprocessing wear make them especially susceptible to the identification errors documented below. George et al. describe a simplified SPD-to-OR cycle [?]:

1. Point-of-use preparation: at the end of a procedure, instruments are prepared for transfer from the OR to sterile processing. This can include gross cleaning, use of sterile water, and enzymatic pretreatment to keep material from drying on instruments.

2. Transport to sterile processing: used instruments are moved from the OR or procedural area to the sterile processing unit.

3. Decontamination: instruments are cleaned and decontaminated. Manual scrubbing, ultrasonic cleaning, and automated washer systems are common parts of this stage.
4. Assembly and inspection: after decontamination, instruments move to the assembly area. Technicians inspect instruments, assemble instrument pans or sets, replace missing items, and place instruments into containers or wraps for sterilization.
5. Sterilization: packaged instruments are sterilized using methods appropriate to the device and material. High-temperature methods such as autoclave sterilization and low-temperature methods such as chemical or radiation-based sterilization are used.
6. Storage: once sterilized, instruments and trays are cooled as needed and placed in storage until they are required again.
7. Return to use: when needed for surgery, stored instruments are obtained and brought back to the OR for the next case.

3.3 Narrowed Problem

Within the full SPD-to-OR cycle, this project narrows to tray reconstruction and verification: the point where cleaned instruments are inspected, identified, matched to count sheets, and rebuilt into procedure-specific trays. Nichol and Saari built a risk model of the full surgical instrument cycle because published error studies had documented that defects occurred but had not systematically mapped where in the instrument lifecycle—from OR use through cleaning, assembly, sterilization, storage, and return—errors were most likely. Their model, which studied reusable stainless steel surgical instruments, decomposed the cycle into 104 human tasks and found that 91% of modeled error risk occurred during the sterile-processing portion, especially the decontamination and assembly-and-inspection steps. All eight highest-risk tasks involved human visualization, sorting, and local tray knowledge [?].

3.4 Stakeholders And Importance

Instrument readiness matters because surgery is time-sensitive, coordination-heavy work. When an instrument set is wrong, incomplete, unclean, damaged, or unavailable, the OR team may need to pause, search for replacement instruments, open backup trays, change the procedure flow, or delay the case [?], [?]. Dirty or damaged instruments add a direct safety concern. A 2024 hospital inspection reported pitting (surface corrosion), stains, sticky residue, rust, scratches, and other concerns in randomly selected ready-for-use trays [?]. The Pennsylvania Patient Safety Authority warns that contaminated instruments can put patients at risk of surgical site infection and can also cause lost OR time if discovered after a procedure has begun [?]. SPD backlogs have escalated beyond individual delays: in 2025, Colorado's largest hospital paused all surgeries for a week after inspectors found pitting (surface corrosion), stains, and residue in randomly selected ready-for-use trays [?]. The stakeholders include:

- Patients, who depend on sterile, functional instruments during surgery.
- Surgeons and OR teams, who need complete and usable sets to maintain procedure flow.
- SPD technicians, who perform detailed, high-volume work under production pressure.
- SPD supervisors and educators, who manage training, quality assurance, staffing, documentation, and feedback loops.
- Hospitals, which lose time and capacity when instrument issues delay cases or require rework.

3.5 General Interest Beyond One Local Practice

Most people will undergo surgery at least once [?]. Each procedure depends on correctly assembled, sterile instrument trays, which means each procedure depends on the same manual identification and reconstruction task that the sections above document as vulnerable. The problem is not confined to one hospital's workflow; it is embedded in every OR that relies on prepared instrument sets.

3.6 Root Cause Analysis

The fishbone head for this analysis is:

> Tray content errors during reconstruction of reusable stainless steel instrument trays.

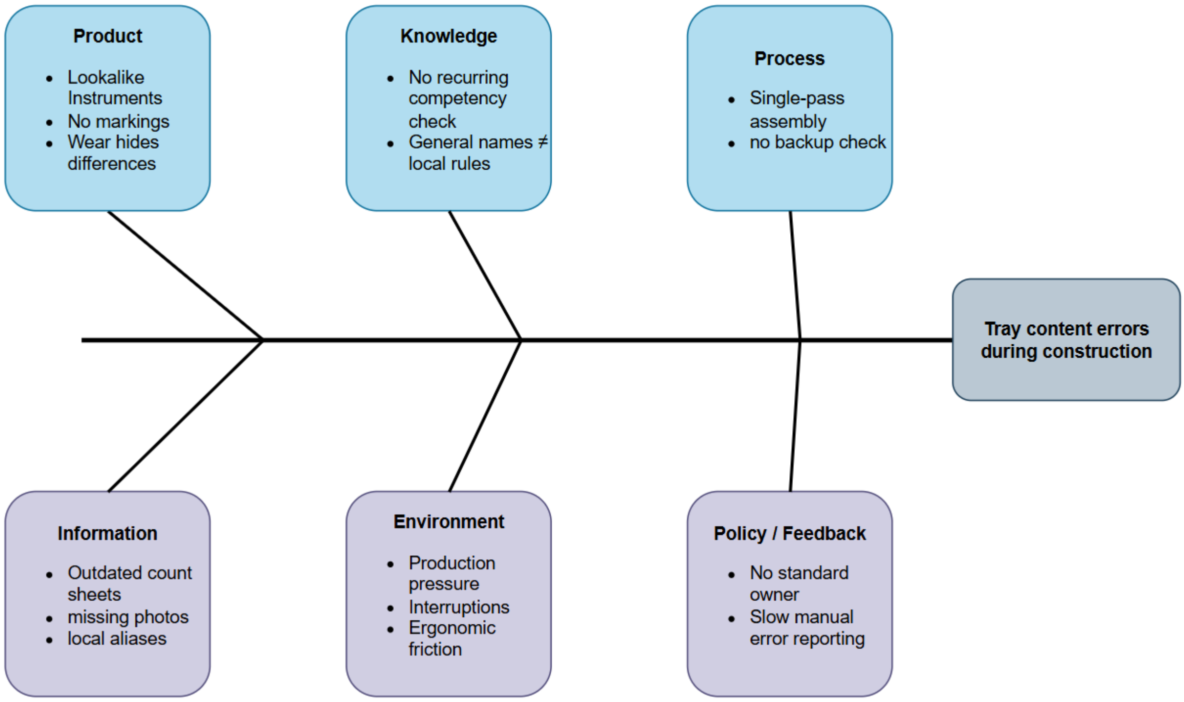


Figure 1: Fishbone root cause analysis of tray content errors during reconstruction of reusable stainless steel instrument trays.

Product.

Same-family instruments (Kelly/Crile, straight/curved Mayo) differ only in subtle dimensional features with no intrinsic labels or markings [?]; [?]; [?]; [?]; [?]; [?]. For example, a tray requiring four 14 cm Kelly clamps and two 16 cm Kelly clamps can be misassembled as two 14 cm and four 16 cm clamps—the total count is correct but the instrument variant is wrong, a pattern that Zhu et al. found accounted for 44.0% of all instrument packaging errors [?]. Steam autoclaving destroys adhesives and fades inks, laser etching is unstandardized across manufacturers, and reprocessing wear further reduces visible cues [?]; [?].

The instrument supply chain evolved around surgical function, not SPD identification. Manufacturers bear labeling cost, and SPD receives the benefit but SPD has no purchasing authority over instrument design [?]. No stakeholder has enough individual incentive to change instrument-marking practices because error costs are small incremental delays spread across the workflow, not a direct impact on surgery or surgical liability [?]. No regulatory body mandates marking because no sentinel event with a clear causal line to patient death has been attributed to unlabeled instruments [?]. Retrofitting across existing inventory is prohibitive.

Tray composition, instrument design, inventory, and unstandardized nomenclature make each tray a complex product to reconstruct reliably [?]. Tray contents are local design decisions rather than fixed universal lists. Procedure-specific, surgeon-specific, and hospital-specific decisions determine which instruments and quantities belong in a set [?]; [?]. A user can know instrument families and still select the wrong item or quantity for a given tray. Reconstruction must also include rejection of visibly unacceptable items, not only selection of named items. Otherwise damaged or suspect instruments can remain in a tray despite correct identity and count [?]; [?]; [?]; [?]; [?]; [?]; [?]; [?].

Knowledge.

Three sub-causes converge. First, knowing the general name is not sufficient. Technicians must see the difference between nearly identical instruments [?]; [?]. Visual discrimination is a perceptual skill that requires deliberate practice with feedback [?]; [?], but the SPD workplace provides no structured opportunity for this practice. Preceptors train on the fly, production pressure prevents deliberate study, and no mechanism identifies which visual discriminations an individual has not yet mastered [?]; [?]. Certification failure rates of 28–50% confirm that the technical demands are substantive, yet the role remains classified and compensated as entry-level support labor [?]; [?]; [?].

Second, general instrument knowledge and local tray rules are separate domains. Manufacturers produce physically different tools under the same generic name [?]. Tray composition is driven by surgeon preference, hospital case mix, and historical accretion rather than a universal standard [?]; [?]. Local aliases accumulate organically on count sheets with no central nomenclature authority [?]. Count sheets encode tray-specific rules that exist only in that hospital's workflow [?]; [?].

Local tray configurations operate at substantial scale. At Aarhus University Hospital, Denmark's largest hospital, a tray-optimization project catalogued 1,340 instrument trays across 12 surgical specialties and 74 operating rooms, averaging 30 instruments per tray [?]. No single tray design generalized across specialties or procedures; the hospital maintained 258 distinct tray types before optimization. A technician rotating between specialties encounters different tray rules, different instrument sets, and different quantities for each room.

Even within a single tray type, usage varies by case and procedure. At Maastricht University Medical Center in the Netherlands, one Major General Surgery tray contained 94 reusable instruments used across 162 multi-disciplinary procedures [?]. On average only 19 instruments were used per case; 10 were never used across all 162 procedures, and the remaining 65 appeared in fewer than 20% of cases. Observation-based optimization removed those rarely used instruments, reducing the tray's size and weight by 54% [?]. A technician cannot rely on frequency of use to know a tray's content.

Third, certification proves a starting point rather than local readiness. Neither HSPA nor CBSPD mandates recurring competency checks tied to local tray knowledge [?]; [?]; [?]; [?]; [?]. Training inconsistency persists because no single role owns the standard curriculum for each tray. Differences across clinical supervisors, shifts, and training tenures produce conflicting mental models of the same tray [?]; [?]; [?]; [?].

Process.

Building a tray forces picking, counting, checking, and arranging into one step with no double-check system. Alfred et al. found that 55.0 percent of tray defects occur during assembly [?]. Nichol et al. found that 88.6 percent of observed errors involve visualization tasks such as inspection, identification, function checking, and sorting [?]. Sequential verification would double labor per tray, and no accreditation mandate from the Joint Commission, CMS, or HSPA requires a two-person check for tray assembly [?]; [?]; [?].

Productivity metrics structurally incentivize single-pass assembly. The AAMI benchmarking framework uses worked-hours-per-unit (WHPU), reinforcing throughput as the primary performance signal

[?]. Instrument damage such as corrosion, stains, burrs, and worn jaw serrations can be missed when inspection relies on rapid visual checks under poor lighting [?]. Workers use only their eyes and hands, without magnification, side-by-side comparison guides, or decision-support tools at the workstation [?]. No worker can remember every item in every tray when a single hospital may manage hundreds of tray configurations, each with 30 to more than 100 instruments. Production pressure and cognitive load cause workers to break from the count sheet and rely on memory, especially for familiar trays [?]. No formal measurement of count-sheet compliance exists, so the gap between assumed behavior and actual behavior is invisible to management. Tray errors have not crossed the sentinel-event threshold needed to trigger regulation [?].

Information.

Count sheets become incomplete or outdated because tray contents change continuously. Instruments are added, removed, substituted, or replaced as surgeon preferences shift, manufacturers discontinue items, or vendors supply new loaners. No dedicated role or budget exists for count-sheet maintenance. Updating a count sheet is a manual, discretionary task that competes with production work. Without a formal change-management workflow, sheets accumulate inaccuracies across multiple revision cycles [?]. Missing or outdated identification information, IFUs, and tray listings at the workstation force workers to rely on memory [?].

Loaner trays introduce a separate information failure. They arrive with manufacturer-stocked contents and vendor documentation that uses the vendor's naming conventions and part numbers, not the hospital's local aliases or count-sheet format. SPD receives the tray with hours or minutes to process it before surgery, and the worker must reconcile the vendor's nomenclature against the local count sheet with no lead time to prepare or cross-reference [?].

Automated counting and tracking technologies such as RFID have been explored to reduce tray-reconstruction errors, but adoption remains limited by infrastructure cost for most hospitals [?, ?, ?, ?]. Sensor-based autonomous verification has also been investigated but requires real-instrument access and deployment validation beyond the current project scope [?, ?].

Environment.

Multiple environmental conditions reduce the attention and time available for tray reconstruction. Pressure and interruptions break focus for detail-heavy visual work [?]; [?]. The open-plan workstation layout makes it difficult to shield assembly workers from interruptions. Phones ring with OR requests, surgeons call for add-on instruments, and runners deliver and collect trays [?]. Uncomfortable workstations, awkward reaches, and poor setup make a job that already needs close attention harder [?]; [?]; [?]. Ergonomic improvements compete for facilities budget with clinical areas that have more organizational visibility. SPD staff have less organizational voice than clinical departments [?].

The wage level is a structural cause that amplifies every other category. SPD is a cost center within the hospital. It generates no direct revenue, so staffing budgets are structurally constrained relative to OR capacity, which is a revenue center [?]; [?]. Hospitals routinely add surgery suites without proportionally increasing SPD headcount [?]. Average SPD pay is $69,217, with a 14 percent drop in respondents receiving raises. SPD workers describe wages as "less than your local Starbucks wage" [?]; [?]. Fewer staff means less time for training and less thorough checking [?]; [?]; [?]; [?]; [?].

Three structural mechanisms—historical classification as materials management, absence of professional licensure, and low union representation—produce this wage level. Together they create a self-reinforcing cycle: structural underinvestment produces low wages, low wages produce chronic vacancies, vacancies increase production pressure, and production pressure preempts the training and process improvements that could reduce errors [?]; [?]; [?]; [?]; [?].

Policy & Feedback.

SPD governance is decentralized across shifts. No single role is responsible for maintaining, auditing, or enforcing count-sheet accuracy or tray-assembly procedure across all shifts [?]; [?]. Creating a single authoritative standard for each tray requires a designated owner, a maintenance process, and enforcement across shifts, but none of these are budgeted.

Certification bodies (HSPA, CBSPD) recommend but do not mandate recurring competency checks tied to local tray knowledge. When competency programs exist, they are locally administered with no external audit [?]; [?]. The absence of a mandate means that even published, evidence-backed training programs — such as the Ofstead et al. (2023) borescope inspection pilot — remain discretionary investments that compete for budget with production headcount, rather than required practice.

Error reporting is manual, slow, and deprioritized. Errors discovered during surgery are documented, if at all, in incident reports that may take weeks to reach SPD management. By that time the specific tray, technician shift, and training gap are hard to identify, so the error teaches no preventive lesson [?]; [?]. The hospital incident reporting system was designed for clinical errors such as medication errors, falls, and wrong-site surgery. Tray-content errors do not fit neatly into the reporting categories, so they are underreported. Without closed-loop tracking that ties an error back to a root cause, downstream detection does not drive upstream improvement [?]; [?].

Structured process-improvement approaches can substantially reduce these defects. Using Six Sigma methodology, Natarus et al. improved first-pass yield from 81.0% to 97.4% and reduced the tray defect rate from 2.2% to less than 0.1% in a hospital sterile-processing department, demonstrating that targeted workflow and training interventions can close the gap described above [?].

Consolidated Root Cause Chain.

Sterile processing evolved from materials management, producing a logistics-oriented workforce without the professional infrastructure of clinical departments. Compensation and professional infrastructure reflect that origin: below current technical demands. Hospital associations resisted mandatory certification because it would create upward wage pressure. Low union representation leaves no institutional mechanism to counter cost-center budget logic. Temporary contract SPD technicians earn 40 to 80 percent more than permanent staff [?]; [?].

Compounding these workforce conditions, the hospital financial structure separates OR revenue from SPD cost: every SPD investment is visible as expense while error reduction has no standard metric. Hospitals add surgery suites without increasing SPD headcount, making underinvestment rational at each hospital because error costs are diffuse workflow friction.

A third structural driver compounds the first two: no information architecture connects OR outcomes to SPD actions, so upstream errors have no downstream feedback.

The pattern is visible in SPD training evidence. Ofstead et al. (2023) demonstrated that structured borescope inspection training for sterile-processing professionals produces large, durable knowledge gains: mean test scores rose from 41% to 84% after the workshop and remained at 90% after two months, and trainees identified actionable defects on patient-ready endoscopes at their own facilities [?]. Despite publication in a peer-reviewed infection-control journal (the *American Journal of Infection Control*) and endorsement by the HSPA Foundation, no mechanism exists to compel or incentivize adoption of this training model. The intervention demonstrates educational efficacy (Kirkpatrick Level 2), but the adoption decision depends on regulatory mandate, liability risk, or visible cost avoidance (Level 4), conditions that the structural drivers above keep absent. This pattern of strong learning evidence with no adoption pathway is the same dynamic that any future training tool, including TrayGuard, must anticipate.

A cross-profession comparison confirms that these structural drivers, not the quality of available evidence, explain the absence of reform. Respiratory therapy followed the canonical professionalization trajectory: the American Association for Respiratory Care (AARC) established a voluntary certification in 1969, unified credentialing under the National Board for Respiratory Care in 1974, and then lobbied state legislatures one by one; California passed the first licensure law in 1982, and by 2004 all 48 contiguous states had enacted licensure for respiratory therapists, enforced by state boards with disciplinary authority [?]. Surgical technology is mid-trajectory: the Association of Surgical Technologists (AST) has driven over a dozen states to require graduation from a CAAHEP-accredited program and Certified Surgical Technologist (CST) credentialing, though many states still lack regulation [?].

Sterile processing followed none of this path, and the reason is not that its technical demands are lower. A comparison of enabling conditions makes the gap explicit:

Table 1: Enabling conditions for professionalization: respiratory therapy, surgical technology, and sterile processing

Enabling condition	Respiratory therapy	Surgical technology	Sterile processing
Visible patient harm creating political will	Respiratory-failure patient deaths drove legislative urgency	Surgical-site infection outbreaks provided the harm narrative	Tray-content errors have not crossed the sentinel-event threshold [?]
Professional association with lobbying infrastructure	AARC maintains full-time government-affairs staff and drafts model legislation [?]	AST runs a sustained state-by-state legislative campaign [?]	HSPA and CBSPD are voluntary certification bodies, not advocacy organizations
Standardized education pipeline	CoARC-accredited programs are the required entry path	CAAHEP-accredited programs required by law in regulated states	No mandated program accreditation; 400 hands-on hours with no curriculum standard
Direct patient-care lineage	Evolved from ICU respiratory support (1950s–60s)	Evolved from intraoperative assistance in the OR	Evolved from central supply and materials management [?]

Each condition maps to a structural driver identified above. The absence of visible harm corresponds to the broken feedback loop that keeps error costs invisible. The absence of lobbying infrastructure and education standardization corresponds to the weak professionalization that followed from SPD's materials-management origin. The absence of a patient-care lineage corresponds to the cost-center classification that suppresses wages and organizational voice. Together these explain why a role with comparable technical demands remains a semi-profession while adjacent roles achieved state-enforced regulation [?].

No information architecture connects downstream OR outcomes to upstream SPD actions. The feedback loop from error to correction is broken, so the same failures repeat without triggering systemic change. Error reporting is designed for clinical events, not tray-content errors.

The system remains stable because no stakeholder has both the incentive and the authority to change it. Manufacturers control instrument design but bear no cost from SPD identification errors. Hospital administrators control SPD budgets but see wage increases as expense and error costs as invisible friction. SPD workers perform the work but lack the union representation, professional licensure, or organizational voice to demand structural change. Regulatory bodies have not acted because tray errors have not crossed the sentinel-event threshold.

4 First Attempt: A Computer-Vision Tray Checker (Matthew, Qiyu)

4.1 CV Literature Context

Computer vision was selected as the initial approach because published work showed that automated surgical-instrument detection and counting could work in experimental settings. Deol et al. trained a deep-learning detector on 1,004 images of 11 instrument categories and reported 98.5% precision and 99.9% recall on static images with real-time inference at 40 FPS [?]. Atabuzzaman et al. built a multi-view CNN-transformer system for 95 instrument subtypes across three surgical trays and demonstrated improved CSSD workflow efficiency in a deployed user study [?]. These papers supported CV as a candidate for tray verification. The YOLO architecture used in these studies originated as a unified real-time detection framework [?], and surgical-instrument datasets continue to expand the class coverage available for training [?].

4.2 Requirements and System Design

Four operational requirements constrained the original CV-based tray-checker design. First, the system had to run entirely on local hardware in the SPD. Hospital IT environments rarely provide GPU-equipped workstations [?, ?], and capital budgets favor sterilization equipment over departmental computing hardware [?]. Second, patient privacy required that tray images never leave the SPD: images could identify specific procedures, surgeons, and OR schedules, and HIPAA-covered entities must evaluate even indirect identifiers before transmitting data off-site [?]. Third, the detection pipeline needed to handle the variable illumination, reflective tray surfaces, and overlapping tool layouts that SPD workflows present. Fourth, the system had to incorporate human judgment for uncertain cases and require explicit final sign-off, supporting rather than replacing the technician.

The local-inference and privacy requirements drove the capture-fixture design. A fixed overhead camera mounted on a portable weighted board preserved repeatable camera geometry across work surfaces without cloud connectivity or GPU infrastructure.

The detection design addressed the robustness requirement through a YOLO object detector with a two-threshold confidence band, drawing on the selective-classification literature in which a model abstains rather than forcing a low-confidence prediction [?]. Detections above a high threshold $T_{\text{present}} = 0.8$ were accepted as present; detections between the low threshold $T_{\text{review}} = 0.3$ and T_{present} triggered a human review state; and the absence of any candidate above T_{review} prompted a rescan or missing-item recommendation. The review band was designed to be wide in order to accommodate the miscalibration that object detectors exhibit near their decision boundary [?]; [?], and the operating point would be refined after training on instrument-specific data. The initial thresholds were chosen to prioritize precision for the "present" decision, following the principle that safety-critical applications require a higher confidence bar than benchmark-default values [?].

These thresholds defined a four-state tray-status machine (Table 2).

The human-in-the-loop and robustness requirements shaped the technician-facing UI (Table 3).

Table 2: Original planned tray-status logic using two-threshold confidence band

User-Facing State	Decision Logic
`Present`	Required class has at least required count of detections above high threshold `T_present`, no active similar-class or image-quality risk flag, and duplicate boxes reconciled.
`Needs Review`	Detections between `T_review` and `T_present`, similar class appears, count unstable, unknown or extra object plausible, or scan quality degraded.
`Missing`	No candidate above low threshold `T_review`, and scan quality acceptable enough that absence is meaningful.
`Rescan Recommended`	System lacks evidence but glare, occlusion, blur, poor framing, or lighting makes a missing label unsafe.

Table 3: Original planned technician UI requirements

Requirement	Design Direction	Rationale
One primary action per state	Show the next meaningful action: start scan, rescan, review issue, confirm tray, or stop.	The technician's decision depends on the tray state; surfacing all controls buries the relevant one [**?**].
Clear tray-status hierarchy	Summarize Complete, Missing, Extra, and Needs Review above detailed detections.	The tray outcome is the priority; detection details support investigation.
Fast correction and rescan	Provide direct correction, dismiss, and rescan paths instead of forcing a workflow restart.	A workflow restart wastes time and discourages correction.
Specific review reasons	Label review prompts with the cause: low confidence, similar-class risk, glare, overlap, unknown object, or count mismatch.	The technician can accept, dismiss, or rescan without guessing the review trigger.
Visual-first feedback with optional sound	Persistent visible status as the source of truth; short optional audio for major state changes.	Visual status provides a continuously available source of truth independent of SPD noise.
Alert restraint	Escalate only conditions that change the tray decision or require user action.	Conditions that do not change the tray decision do not warrant escalation.
Human sign-off	Require explicit final confirmation. The system supports the decision, not silently makes it.	The system supports the decision, not silently makes it [**?**].
Manual fallback	Make it clear how to continue if camera, model, or scan quality is unavailable.	Tray processing cannot halt when the capture fixture, model, or scan is unavailable.

4.3 CV Experiments and Results

The published literature established that deep-learning object detectors can recognize surgical instruments with high accuracy in controlled settings [**?**, **?**]. Those results were obtained under fixed lighting, non-reflective surfaces, and separated instrument layouts—conditions that do not reflect the variability of real SPD workflows.

Domain shifts in illumination, background appearance, and object arrangement degrade detection even for models that perform well on matched benchmarks [?, ?]. Specular reflections from metallic surfaces further reduce accuracy [?, ?, ?], and object detectors tested under lighting perturbations show significant performance loss [?]. No prior study had systematically measured how these factors interact for surgical-instrument detection in an SPD-like setting.

We designed staged experiments testing detection robustness along three axes directly relevant to SPD conditions: illumination (800–80 lumens), surface reflectivity (matte vs. reflective backgrounds), and layout occlusion (separated vs. overlapping). All experiments used YOLO11s on a controlled self-collected dataset with five classes (four spoon variants and a fork) arranged in a fixed 11-step brightness ladder across two background conditions and two layout conditions. Each stage contained 3–43 training images (47–109 total images per stage). The design isolated each variable independently rather than mixing effects. Published studies report mAP50 exceeding 0.99 for surgical-instrument detection under controlled lighting and layout [?, ?]; our experiments test whether that accuracy holds under SPD-relevant variations.

Architecture selection. The staged experiments all used YOLO11s at 640×640 px. This configuration was selected through a controlled architecture sweep on the Lavado surgical-instrument dataset. We evaluated four YOLO11 variants (n, s, m, l) at 640 px and selected variants at 960 px alongside RT-DETR-l, a Transformer-based real-time detector [?], to assess the accuracy-latency tradeoff at higher input resolution (Table 4). YOLO11s at 640 px was chosen for the remainder of the experiments: it achieved the best balance of mAP50-95 (0.928) and latency (4.2 ms), and higher-resolution variants offered marginal accuracy gains at substantially higher compute cost.

Table 4: Architecture sweep on Lavado dataset

Model	Precision	Recall	mAP50	mAP50-95	Latency (ms)
YOLO11n (640px)	0.936	0.939	0.980	0.914	2.4
YOLO11s (640px)	0.958	0.944	0.986	0.928	4.2
YOLO11s (960px)	0.965	0.968	0.990	0.948	7.1
YOLO11m (640px)	0.979	0.952	0.991	0.940	6.8
YOLO11m (960px)	0.974	0.958	0.991	0.945	14.6
YOLO11l (640px)	0.964	0.949	0.988	0.932	8.0
YOLO11l (960px)	0.963	0.960	0.990	0.940	17.8
RT-DETR-l (640px)	0.993	0.991	0.991	0.957	12.4

Lavado baseline. A YOLO11s model was also trained on the Lavado surgical-instrument dataset: 2,106 training images across four real instrument classes (scalpel, straight dissection clamp, straight Mayo scissors, curved Mayo scissors). After 100 epochs the model achieved precision of 0.972, recall of 0.978, mAP50 of 0.991, and mAP50-95 of 0.945 on the held-out test set (302 images). Per-class AP50/AP50-95: Scalpel n4 (0.995/0.929), Straight Dissection Clamp (0.983/0.892), Straight Mayo Scissor (0.993/0.977), Curved Mayo Scissor (0.994/0.982). This established that the training pipeline and architecture perform competitively on a published surgical-instrument dataset, consistent with prior work on CNN-based surgical instrument recognition [?], before the staged proxy experiments.

Brightness degradation. Two experiments measured monotonic mAP50-95 decay as the train–test illumination gap widened. `brightest_train_darker_test` trained on 800-lumen separated images and tested across all dimmer levels, producing mAP50-95 of 0.815. `darkest_train_brighter_test` trained on 80-lumen separated images and tested across all brighter levels, producing mAP50-95 of 0.822. The decay was asymmetric: the brightest-trained model retained higher recall under moderate drops, while the darkest-trained model recovered more quickly as light increased. Per-brightness metrics confirmed the trend was consistent across both matte and reflective backgrounds.

Background transfer. Two experiments tested whether the model learned object shape or overfit to the scene background. `matte_train_reflective_test` trained on matte-background images and tested on reflective-background images across all brightness levels, producing mAP50-95 of 0.776. `reflective_train_matte_test` trained on reflective-background images and tested on matte-background images, producing mAP50-95 of 0.826. The reflective-to-matte direction was stronger, suggesting that reflective training data provides broader feature invariance.

Occlusion stress. The separated-to-overlay layout shift produced the largest degradation. `separated_train_overlay_` trained on non-overlapping layouts and tested on overlapping layouts, producing mAP50-95 of 0.478 compared with 0.995 on the matched-condition baseline. This gap was consistent across all brightness ranks and both backgrounds, showing that training on neatly arranged instruments does not transfer to cluttered arrangements. The overlay condition was the only experiment where recall fell below 0.70, making it the clearest product boundary: the system should request a rescan whenever tools overlap.

Per-image analysis confirms the failure pattern: in the overlay condition on matte backgrounds, the model detects 2 of 5 instruments on average (recall 0.4) with zero false positives across all brightness levels. On reflective backgrounds, recall rises to 0.8 with occasional false positives. The dominant failure mode is systematic under-detection rather than spurious extra predictions.

Table 5: Aggregate test-set metrics for the seven staged brightness experiments

Stage	Precision	Recall	mAP50	mAP50-95
`brightest_train_darker_test`	0.898	0.953	0.993	0.815
`darkest_train_brighter_test`	0.955	1.000	0.984	0.822
`bright_train_dim_test`	0.992	0.992	0.995	0.904
`dim_train_bright_test`	0.907	0.851	0.995	0.886
`matte_train_reflective_test`	0.915	0.904	0.957	0.776
`reflective_train_matte_test`	0.832	0.893	0.975	0.826
`separated_train_overlay_test` (overlay)	0.836	0.671	0.716	0.478
`separated_train_overlay_test` (baseline)	0.998	1.000	0.995	0.995

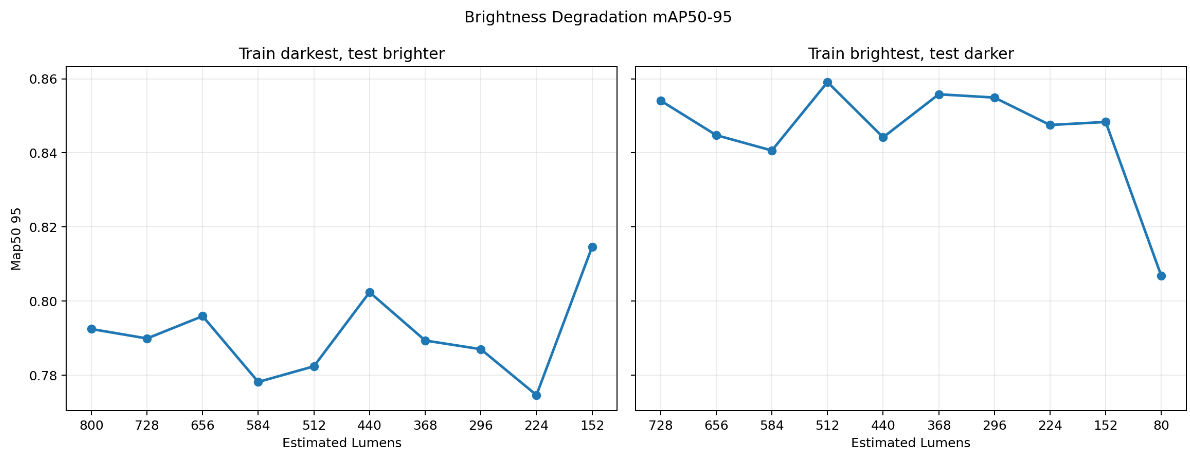


Figure 2: Brightness degradation mAP50-95 across the estimated-lumens range. Left panel: model trained on darkest (80 lumens), tested on brighter levels. Right panel: model trained on brightest (800 lumens), tested on dimmer levels.

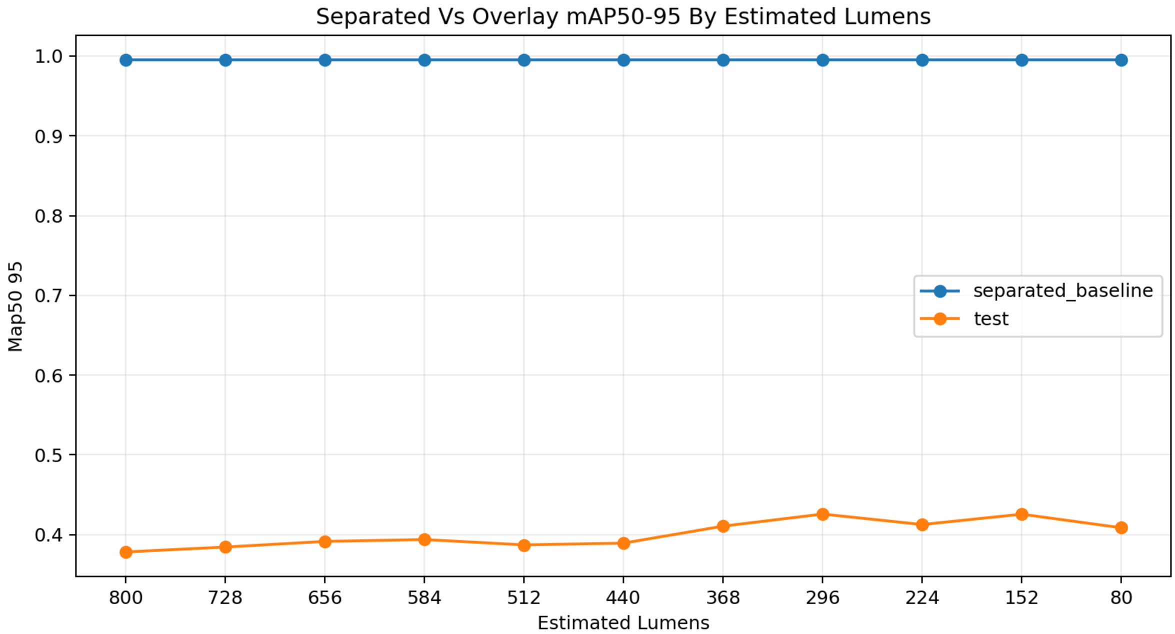


Figure 3: Separated versus overlay layout: mAP50-95 across the estimated-lumens range. The overlay condition produces the largest degradation (mAP50-95 = 0.478).

Real-instrument experiments. The brightness experiments above used a spoon proxy to measure degradation under controlled variables. A further set of experiments tested detection on real surgical instruments—six classes (scissor1–4, forcep, scalpel) from a surgical training kit photographed on matte and reflective backgrounds across four sessions. Three staged experiments were trained for 100 epochs each using YOLO11s and validated on held-out sessions:

Table 6: Aggregate test-set metrics for the three real-instrument experiments

Stage	mAP50	mAP50-95	P	R
`matte_train_reflective_test`	0.336	0.233	0.425	0.236
`reflective_train_matte_test`	0.160	0.119	0.123	0.160
`shape_similarity_real_proxy`	0.300	0.253	0.239	0.347

These scores are substantially lower than the spoon-proxy results. Per-class analysis (Table 38) reveals a systematic pattern: each image contains exactly six instruments in fixed positions per session, so a model can learn which classes appear at which image locations rather than learning visual features. Different classes survive under different splits—scissor1 reaches 0.862 AP under the shape-similarity split but falls to 0.000 under lighting transfer splits, while forcep and scissor2 follow the opposite pattern. Only scalpel, the most visually distinct instrument (single blade vs. multi-loop scissors), transfers across conditions (0.717 AP in reflective-to-matte). The confusion matrix (Figure 9) confirms that missed detections are the dominant failure mode: when the model fails, it produces a false negative rather than a false positive. The lighting transfer experiments are therefore confounded by position-cue leakage: the test metric reflects both lighting difficulty and layout generalization. These results motivated the pivot to a training platform where controlled physical cards replace uncontrolled instrument recognition.

A separate experiment confirmed that adding lighting variety during training improves held-out-condition detection (Appendix A.3).

Lessons learned. Detection worked reliably in controlled, matched-condition settings and degraded under every variation that SPD workflows present—variable illumination (mAP50-95 as low as 0.822), reflective tray surfaces (mAP50-95 as low as 0.776), and overlapping instruments (mAP50-95 = 0.478). The real-instrument experiments were confounded by position-cue leakage and failed to transfer across lighting conditions. These results showed that a CV-only tray checker could not deliver reliable detection under SPD conditions within the project's timeline and resources. Controlled lighting and layout are prerequisites for reliable detection. The training-centered pivot is the subject of Section 5.

4.4 CV Requirements Assessment and Gap Analysis

The original tray-verification concept defined requirements for automated tray-status logic, technician-facing UI, physical capture fixture, and detection reliability. Each requirement is assessed below.

Table 7: CV design outcome versus requirements

Requirement	Target	Achievement	Gap
Tray-status logic with two-threshold confidence band	Present, Needs Review, Missing, Rescan states	Logic specified and documented	Not tested with real SPD workflow data
Technician-facing UI with one primary action per state	Clear hierarchy, fast correction, specific review reasons, human sign-off	UI requirements specified	Not implemented beyond design documentation
Physical capture fixture	Overhead camera on portable weighted board	Fixture design documented	Not fabricated or tested in SPD conditions
Lighting robustness	Detection maintains mAP50 above 0.7 across SPD-like illumination range	Evaluated across 800–80 lumens; overlay layout caused largest drop	mAP50-95=0.478 in overlay condition; lighting still a deployment barrier
Lookalike discrimination	Correctly distinguish straight/curved, narrow/broad same-family pairs	Evaluated with 6 real instrument classes (scissor1–4, forcep, scalpel)	Per-class AP varies by split from 0.000 to 0.862; position cues dominate over shape; only visually distinct scalpel transfers reliably
Open-set rejection	Detect unknown objects as unfamiliar rather than assigning spurious labels	Not evaluated — no distractor dataset available; open-set recognition methods exist but require threshold calibration [?, ?]	
Synthetic-to-real transfer	Synthetic pretraining enables few-shot real deployment	Not evaluated — no 3D instrument models available for synthetic data generation; few-shot detection approaches [?] could reduce annotation burden	

The training-centered pivot (Section 5) preserved the project's technical contributions (the CV experiments, data collection pipeline, and detection code are documented in Section 4) while replacing the main evidence path with a claim that was buildable, testable, and interpretable within the project's resources. The CV infrastructure remains available as a future support layer after separate validation

(Section 4.3).

4.5 CV Implementation Summary

The following CV subsystems were implemented:

- **Data collection and annotation pipeline.** A webcam capture tool with bounding-box annotation UI supports rapid multi-session data collection with per-class image output, YOLO label export, and session save/load (`src/trayguard/data_collection/`).
- **Training infrastructure and benchmark harness.** A Hydra-configured training pipeline supports model, dataset, and hyperparameter overrides across YOLO11n/s/m/l variants and RT-DETR-l, with a multi-variant benchmark for accuracy–latency comparison (`src/trayguard/training/`).
- **Legacy tray-check demo.** A live YOLO tray-check interface demonstrates camera capture, detection, and tray-status comparison for proof-of-concept review (`src/trayguard/app/streamlit_app.py`).

4.6 Limitations of the CV Experiments

The CV experiments have several constraints that limit the strength of their conclusions. First, the brightness and occlusion experiments used a spoon proxy rather than real surgical instruments. While the proxy controlled for confounds such as shape and reflectivity variation, it does not capture the fine-grained visual differences between functionally similar instrument classes that fine-grained classification methods address [?, ?, ?], and the degradation magnitudes may differ from those of real instruments under identical conditions. Second, the real- instrument experiments were confounded by position-cue leakage: the testing layout used fixed positions per session, so the model learned spatial correlations rather than visual features alone, making the lighting-transfer results difficult to interpret (Section 4.3). Third, all experiments used a single architecture (YOLO11s) and a single training recipe. The architecture sweep (Table 4) identified YOLO11s as the best accuracy–latency balance from eight configurations, but the broader conclusions rely on a single detection architecture; other detectors, pretraining strategies, or data-augmentation pipelines may produce different degradation patterns, and ML reproducibility concerns [?, ?] apply to any single-site evaluation. Fourth, the test conditions—staged brightness ladders, fixed layouts, and controlled backgrounds—do not reproduce the full variability of real SPD workflows, including arbitrary instrument arrangements, occlusion challenges documented in the detection literature [?], open-set distractors, dirt or wear, and operator-induced framing variation. Fifth, the Lavado baseline and real-instrument experiments covered at most six classes, whereas a deployment-ready system would need to distinguish hundreds of instrument types across multiple manufacturers. The evidence supports bounded claims about degradation patterns in proxy settings.

5 Limitations of the Computer-Vision Approach (Matthew)

Computer vision for surgical-instrument verification must generalize across variable lighting, reflective surfaces, overlapping instruments, and instrument variants from different manufacturers at each deployment site. Domain generalization in medical imaging is a well-documented open problem: models trained at one institution routinely degrade when deployed at another due to differences in equipment, imaging conditions, and patient populations [?]. For surgical instruments specifically, visual variability from overlapping tools, reflections, and vendor-specific designs makes generalization across unseen instruments and deployment conditions insufficient with supervised approaches alone [?]. Synthetic data generation can improve model generalization by creating diverse training images from 3D instrument models, but this requires access to detailed CAD files that manufacturers treat as proprietary intellectual property, making the approach infeasible across the full instrument inventory at each site [?]. The number of distinct instrument types, the frequency of new instruments entering circulation, and the fine-grained visual differences between functionally different tools add further complexity.

Another barrier to adoption is the site-specific data collection burden. A typical hospital maintains tens of thousands of surgical instruments organized into thousands of distinct tray types. Collecting and annotating training images for each instrument across the range of lighting conditions, angles, and wear states found in SPD workflow would require thousands of person-hours of expert annotation per site. The annotation and expert-review cost alone would reach tens of thousands of dollars per hospital, and every deployment site must repeat the full cycle independently. A deployment-grade CV tray checker would therefore require site-specific training data, a closed-set operating assumption, human-in-the-loop confirmation, and continuous revalidation as instruments change—requirements that the project cannot satisfy within its timeline and resources.

6 Literature And Related Work (Matthew, Andy, Qiyu)

The CV-first attempt did not produce a deployable verification system within the project's constraints. Returning to the root causes identified in the RCA (Section 3.6), the team revisited the solution space and identified seven intervention points. The following literature examines each relevant area and supports the selected path.

6.1 The Solution Landscape

The solution space for reducing tray-content errors during reconstruction can be organized into seven categories by intervention point. Each category targets a different subset of the root causes, and each faces a different feasibility constraint for this project.

The seven identified solution categories are:

1. **Automated content verification**: post-assembly inspection using CV, RFID, or barcode scanning to verify tray contents.
2. **Physical error-proofing**: modifying the tray or instrument interface (custom foam, color coding) to prevent selection errors.
3. **Workflow / process design**: restructuring task handoffs and verification steps through dual checks or gating sign-offs.
4. **Decision support at point of assembly**: providing standardized count sheets, photos, and alias lookup at the workstation.
5. **Task simplification**: rationalizing tray contents and standardizing instrument sets across procedures.
6. **Workforce / accountability systems**: competency reassessment, closed-loop error tracking, and structured mentorship.
7. **Training and simulated practice**: building local tray skills through retrieval practice, simulation, and feedback before supervised work.

Table 8: Seven solution categories compared by constraint

Category	Intervention point	Example solutions	Root causes addressed	Key constraint for this project
Automated content verification	Post-assembly inspection	CV tray checker, RFID tray scan, barcode line scan, weight check	Product: wear hides differences and Process: no backup check exist	Requires real instruments, lighting, workflow integration; cannot validate without SPD deployment
Physical error-proofing	The tray or instrument interface	Custom foam cutouts, color-coded handles, segregated tray zones	Knowledge: workers must memorize tiny visual details and Process: assembly lacks a separation of pick from check	Expensive custom fabrication per tray; difficult to validate and maintain across tray variants
Workflow / process design	Task structure and handoffs	Dual verification, gating sign-offs, pre-sorted pick lists	Process: no structured backup check during assembly and Policy: nobody formally owns the standard	Primarily organizational change; no technical prototype to build or measure in a controlled study
Decision support at point of assembly	Information available during re-construction	Standardized count sheets with photos, alias lookup, reference kiosk	Information: outdated count sheets force guessing and Knowledge: general instrument names do not convey local tray rules	Improves the reference but does not exercise or measure the technician's own ability
Task simplification	The tray definition itself	Tray rationalization, instrument consolidation, procedure-level standardization	Knowledge: differentiating lookalikes requires local familiarity and Environment: pressure and interruptions break focus	Requires hospital utilization data and multi-stakeholder buy-in; no buildable prototype
Workforce / accountability systems	Competency management and error feedback	Periodic competency reassessment, closed-loop error tracking, structured mentorship	Policy: standards are only tested at hiring and Feedback: slow error reporting lets mistakes repeat	Primarily organizational policy change; no measurable prototype within our project scope
Training and simulated practice	Pre-work and between-work skill building	VR simulation, digital flashcards, tangible tray simulation, structured OJT	Knowledge: lookalike discrimination requires repeated practice and Local knowledge: tray-specific names and rules are not taught in general curricula	Requires learning-science justification and a baseline study, both accessible without SPD deployment

Each category reduces a real risk. Categories 1 through 6, however, all depend on resources that this project cannot access within our project timeline: SPD deployment sites, real instrument inventories, hospital utilization data, institutional policy authority, or multi-stakeholder buy-in. Category 7 – training and simulated practice – is the only category whose primary evidence can be collected with accessible participants (novices), low-cost materials (printed cards), and a bounded pre/post study design on a simulated local tray task. The following sections show why existing commercial approaches also fail to close the gap and then examine the evidence supporting the training path.

6.2 Why Existing Approaches Cannot Close the Gap

Commercial tools in the same categories share the same constraints.

Table 9: Comparable tools and gap relative to TrayGuard

Tool or approach	Target user	Relevant features	Evidence strength	Gap relative to TrayGuard
Tray Pacer	SPD teams and managers	Count sheets, tray assembly support, photos, productivity and proficiency metrics.	Commercial product evidence.	Supports operations, but public materials do not establish the specific pre/post local tray learning study proposed here [?].
LayerJot SID	SPD and surgical instrument teams	Instrument documentation, photos, count sheets, and tray information.	Commercial product evidence.	Strong local-information precedent, but not a demonstrated tag-card learning loop [?].
CensiTrac and similar tracking systems	Hospitals, SPD, OR inventory teams	Instrument and tray tracking, traceability, inventory visibility.	Commercial and operational precedent.	Addresses tracking and lifecycle visibility more than low-cost simulated learning and weak-item training [?].
Touch Surgery and surgical simulators	Medical trainees and clinicians	Repeated simulation attempts, performance logging, skill practice.	Published evaluation and commercial adoption.	Supports simulation logic, but focuses on surgical procedure skills rather than SPD tray reconstruction [?]; [?].
NeuroVase-style tangible cards	Health-professions learners	Physical cue cards, mobile AR, structured curriculum, pre/post testing, usability measures.	Emerging preprint evidence.	Strong design precedent for tangible medical learning, but not sterile-processing content [?].
SPD workstation vendors (Getinge, Skytron, Southwest)	SPD workflow	Assembly workstations, ergonomic support, sterile processing layout.	Commercial product evidence.	Provides ergonomic CSSD infrastructure but no training or assessment loop [?, ?, ?, ?].
SteelcoBelimed SUIS	SPD teams	Vision-based instrument recognition and counting.	Commercial product evidence.	Addresses instrument identification for tracking, but not pre/post learning measurement [?].
AAOS VR CME (Fundamental Surgery)	Orthopedic surgeons	VR simulation with CME accreditation.	Published CME adoption precedent.	Demonstrates that VR simulation can meet regulatory training standards, supporting the broader simulation-credibility argument [?].

6.3 Training and Simulation Precedents

The recurring constraint across SPD training approaches is that hands-on practice and clinical supervisor attention are the scarce resources. Several program examples show the gap between classroom preparation and clinical capacity. Penn Foster's online sterile processing program prepares learners for the CRCST exam but does not require the exam or the 400 hands-on hours [?]. MedCerts states that full CRCST certification requires 400 hours of hands-on experience following provisional certification [?]. AIMS Education requires a 400-hour clinical internship, but its schedules depend on clinical-site availability [?]. Central Sterilization Solutions states its classroom course has no hands-on training in the classroom and that hands-on training is conducted during externship [?]. These examples illustrate the training-access gap: repeatable, feedback-rich practice with local instruments and tray rules is hard to scale before and during real SPD experience because it depends on site access, clinical scheduling, clinical supervisor attention, and available teaching materials.

Structured training programs for sterile-processing staff have shown positive results. Hu et al. developed a CSSD training program using action research; after two rounds of training, CSSD nurses showed significant improvement in professional knowledge [?]. Huang et al. similarly concluded that an action-research-based training program improved CSSD nurses' professional knowledge and skill [?]. Broader allied-health placement research supports treating clinical placements as capacity-constrained environments. A Queensland study found that placement-offer growth did not keep pace with student-program growth [?]. A 2025 Greater Sacramento report identified shortage of clinical placement opportunities as a barrier to workforce pipeline growth [?].

Simulation-based health-professions education with deliberate practice improves knowledge, skills, and behaviors compared with no intervention and outperforms traditional clinical education in meta-analysis [?]; [?]. Effective simulation includes feedback, repetitive practice, curriculum integration, and measurable outcomes [?]. Retrieval practice produces superior long-term retention compared with elaborative study methods such as concept mapping [?], and adaptive spaced education has been validated across healthcare domains [?]. Virtual patient simulation and virtual reality training improve clinical reasoning across health professions [?, ?, ?, ?]. Tangible augmented-reality markers for anatomy learning demonstrate that physical card-based approaches improve spatial understanding [?].

For sterile processing specifically, Ofstead et al. provide the closest direct precedent. Their borescope/endoscope visual-inspection training pilot used a pre-test, structured teaching, hands-on practice, image-based testing, confidence/satisfaction measures, workplace homework, and a delayed booster with certified sterile-processing professionals. Mean test scores improved from 41% at pre-test to 84% after the workshop, and remained at 90% after two months [?].

Barrison et al.'s 2025 scoping review found that electronic flashcards are widely used in health-professions education and that the research base grew rapidly from 2019 to 2024, while cautioning that development and delivery methods are less systematically studied [?]. QR codes on physical models have been used to link training content to digital resources in anatomy education [?]. Meta-analytic evidence shows that organizational training improves learning, behavior, and operational results when it includes practice and feedback [?]. Salas et al. argue that effective training should be treated as a system: analyze the work, design instruction around target competencies, provide practice and feedback, and evaluate transfer [?].

Medical training software already uses the same design pattern of simulation, guided practice, feedback, scoring, and progress tracking. Touch Surgery provides mobile surgical procedure simulations with repeated attempts and scoring, and a validation study found that users improved across repeated attempts while higher-experience cohorts outperformed lower-experience cohorts [?]. Osso VR demonstrated in a randomized trial that VR-trained novices completed a downstream orthopedic procedure more often, made fewer incorrect steps, and finished faster than guide-only learners [?]. Body Interact uses screen-based virtual patient scenarios with immediate feedback, auto-grading, and educator analytics [?]. Surgical Science and Simbionix provide commercial medical simulators across specialties, positioning simulation-based assessment as a credible healthcare training category [?]. These precedents support treating TrayGuard as a health-professions simulation tool that applies established learning mechanisms to a new domain: local SPD tray reconstruction.

No existing study tests whether card-based tray training transfers to real-world SPD assembly. The closest precedents involve procedural skills rather than visual identification. Masoomi et al. (2021) reviewed 121 simulation-based training studies and found that even simpler training models that do not replicate full workplace realism still produced learning that transferred to real clinical settings; transfer did not depend on physical fidelity alone [?]. Tanaka and Curran (2005) found that training participants to identify objects at a specific, fine-grained level (e.g., "great blue heron" rather than "bird") transferred to recognizing new examples and new categories, whereas training at a broad categorical level did not [?]. TrayGuard's card-based module trains the same fine-grained skill: identifying a specific instrument model and its tray position, not just "clamp" or "retractor." A future experiment comparing card-trained and untrained novices on real tray assembly would measure the transfer effect directly.

7 Requirements Definition (Matthew, Andy, Jacob)

7.1 Element Definition

Table 10: Element definition for the TrayGuard system

Element	Definition For TrayGuard
Intended practice	Competency maintenance for local tray reconstruction for technicians who hold certification but need local tray familiarity.
Other practice affected	Instructor review, clinical supervisor coaching, tray content updates, weak-item remediation, and future CV-supported tray review.
Artifact	A local training app, file-backed tray module, printable marker cards, tray-sorting surface, scoring engine, and export/report package.
Problem addressed	Certified technicians need repeated, measurable practice applying local tray rules, counts, aliases, and lookalike discrimination that their certification did not cover.
Technology	Local web/app workflow, JSON tray modules, QR or AprilTag cards, camera or manual selection input, scoring logic, and CSV/JSON export.
Primary users	SPD technicians, trainees, instructors, SPD educators, and future project teams.
Perception goal	The system should feel like a serious practice tool, not a clinical approval device or punitive surveillance system.
Environment	Classroom, skills lab, project demo, or supervised training setting; not a live sterile field or production SPD deployment.
Core function	Convert a local tray module into study, quiz, simulated sorting, feedback, assessment, and learner metrics.
Required behavior	Separate practice from assessment, suppress hints during pre/post modes, classify errors consistently, and export comparable rows.
Structure	Content module, learning workflow, marker/manual input, scoring engine, feedback engine, reporting/export, and instructor review.
Intended effects	Improve simulated tray-sorting accuracy, reduce severe errors, expose weak items, calibrate confidence, and reduce early clinical supervisor burden.
Possible side effects	Overclaiming simulated scores, punitive use of learner metrics, answer leakage, inequitable access to updated modules, and reduced attention to supervised hands-on practice.

7.2 Objective Tree

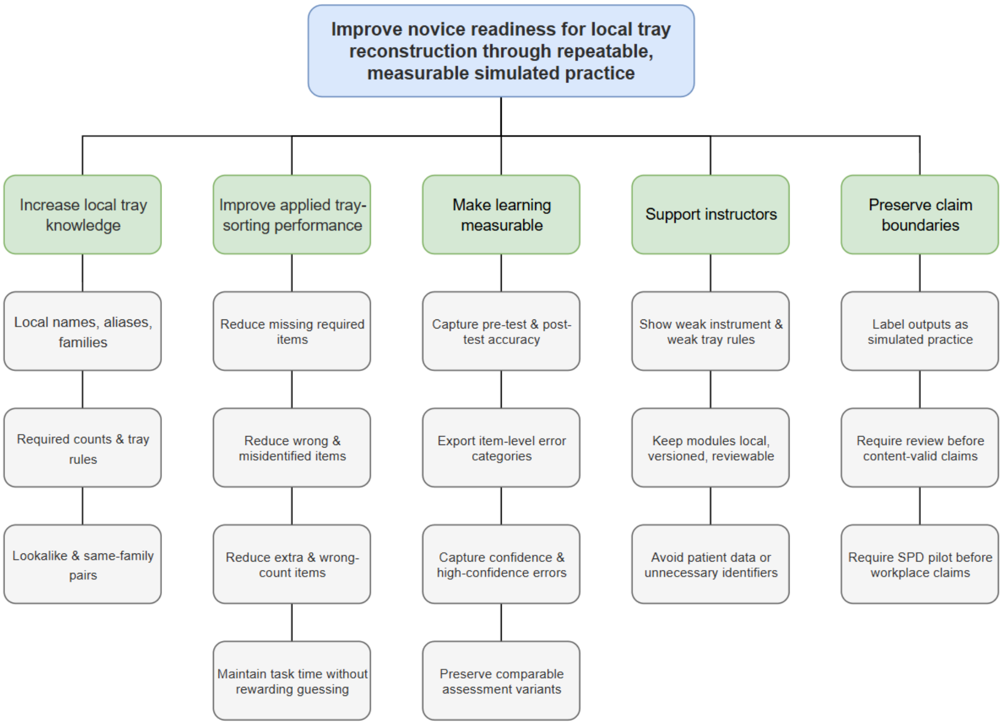


Figure 4: Objective tree for TrayGuard's training intervention.

7.3 Performance Specifications

The first build should use the following targets. They are deliberately formative rather than clinical: they define when the prototype is ready to support a project training-effectiveness claim.

Table 11: Performance specifications and acceptance targets

Area	Target	Evidence Basis	Measurement
Participant count	Minimum 8 baseline participants; target 12 complete participants if schedule allows.	NeuroVase used a 40-participant controlled study for an educational card/AR system, while qualitative methods literature supports small, narrow, homogeneous samples for early formative work [?]; [?]; [?].	Unique participants with complete pre/post attempts.
Completion rate	At least 7 of 8 participants, or 85% when n > 8, complete the full loop without blocking moderator intervention.	Usability must be separated from learning; NeuroVase used SUS and user-experience measures alongside knowledge testing. The System Usability Scale provides a standardized tool for assessing prototype usability in health-education contexts [?]. Additional measures such as the NASA Task Load Index [?] and established usability heuristics [?, ?] could supplement formative evaluation. Health-specific usability scales offer clinically relevant assessment for future SPD pilots [?].	Completion flag and blocking help requests.
Accuracy gain	Mean paired tray accuracy improves by at least 20 percentage points from pre-test to post-test.	Ofstead's SP training pilot showed large pre/post score movement after structured teaching and hands-on practice; TrayGuard should set a smaller but visible formative threshold [?].	`post_accuracy - pre_accuracy`.
Post-test floor	Mean post-test accuracy is at least 80%, and no more than one participant scores below 70%.	The goal is not only improvement from a low baseline; learners should reach a usable simulated familiarity level before the module is considered ready.	Post-test `accuracy_score`.
Severe-error reduction	Mean `missing + wrong + misidentified` errors decrease by at least 30%.	Alfred et al. and Nichol et al. show missing, wrong, extra, and assembly-related errors are operationally meaningful categories [?]; [?].	Paired error-category counts.
Speed without unsafe guessing	Median post-test duration is at least 10% lower, or is no more than	Time matters because tray assembly is workflow-bound, but	`duration_seconds`, paired by learner.

7.4 Quality Function Deployment

Compact QFD:

Table 12: Quality function deployment: needs to engineering targets

Stakeholder need	Engineering characteristic	Target value	Verification
Learners can practice local tray content repeatedly.	File-backed module with required items, distractors, aliases, features, counts, and variants.	One complete `basic_general_tray_v1` module with 8-12 required concepts and 3-6 distractors.	Module schema review and demo load.
Practice feels connected to tray reconstruction, not isolated memorization.	Simulated tray sort using physical cards or manual selection.	Pre-test, practice, post-test, and optional delayed variant all use the same tray rule.	End-to-end run.
Educators can see what learners missed.	Attempt-level and item-level export.	100% of completed runs export all required fields.	Schema validation.
Errors map to operationally meaningful categories.	Scoring engine with `missing`, `extra`, `wrong`, `misidentified`, and `wrong_count`.	Every submitted tray receives category counts.	Unit tests or scored example cases.
Assessment is not contaminated by coaching.	Mode boundaries and feedback suppression.	No hints or corrective feedback in pre-test, post-test, or delayed modes.	UI walkthrough and export mode check.
The physical marker layer does not corrupt scores.	Marker reliability and user confirmation.	At least 95% detection in normal layouts; confirmation before scoring.	Study 0 marker log.
Learner confidence is visible.	Confidence or `Not sure` capture.	High-confidence errors and low-confidence correct answers exported.	Export inspection.
The system remains ethical and nonpunitive.	Privacy and report labels.	Hashed learner IDs, no patient data, and report language that says practice evidence.	Report review.
Future teams can extend the design.	Versioned data model and traceability.	Module version, source notes, verification status, and assessment variant IDs included.	Data file review.

7.5 Function Analysis – Transparent Box Model

The training system is modeled as a transparent box that transforms a local tray module and physical instrument cards into learner performance evidence through sequential subfunctions [?]:

Table 13: Transparent box model for the training system

Input	Subfunction	Output
Local tray module (JSON)	Authoring and verification	Verified tray template with required items, distractors, lookalike pairs, and assessment variants
Verified tray template	Pre-test assessment	Baseline accuracy, error-category counts, and confidence distribution
Instrument cards and tray template	Retrieval-first study	Prompt-before-reveal card practice with distinguishing features and aliases
Study session output	Identification quiz	Quiz accuracy, duration, and item-level confidence per instrument concept
Cards and tray template	Simulated tray sorting	Selected instrument set scored against required counts
Scored tray selection	Feedback generation	Missing, extra, wrong, misidentified, and wrong-count error report
Pre-test and post-test scores	Metrics aggregation	Pre/post accuracy gain, error reduction, confidence calibration, and weak-item report
Aggregated metrics	Export and reporting	CSV/JSON export with attempt-level and item-level records

7.6 Morphological Chart

The morphological chart generates solution alternatives for each subfunction identified in the transparent box model [**?**, **?**].

Table 14: Morphological chart: solution alternatives by subfunction

Subfunction	Alternative 1	Alternative 2	Alternative 3	Alternative 4
Content delivery	File-backed JSON module	Online database	Spreadsheet import	Hard-coded tray lists
Instrument representation	Physical marker cards	On-screen instrument icons	Real instruments	Text-based item list
Marker technology	AprilTag	QR code	ArUco	Manual selection
Learner interaction	Study + quiz + tray sort	Study-only	Quiz-only	Tray-sort only
Assessment mode	Pre/post with matched variants	Single post-test	Pre/post without variants	Continuous assessment
Feedback mechanism	Error-category report	Pass/fail only	Item-level hints	Instructor review only
Scoring approach	Category-based scoring (missing, extra, wrong, misidentified, wrong-count)	Simple accuracy (correct/total)	Weighted error severity	Confidence-adjusted scoring
Export format	CSV + JSON per run	Single PDF report	Instructor dashboard	Raw database access

The selected combination is: file-backed JSON module, physical AprilTag marker cards, full study-quiz-sort learning loop, pre/post matched-variant assessment, category-based scoring, error-category feedback, and CSV/JSON export.

7.7 Weighted Objective Method

The weighted objective method scores each candidate solution combination against weighted criteria derived from the objective tree and stakeholder needs [**?**].

Table 15: Weighted objective method: solution path selection

Criterion	Weight	Training path	CV tray checker	Software-only quiz	Real-instrument CV
Buildability within project scope	0.25	5	1	4	1
Measurable learning evidence	0.20	5	2	3	3
Physical task fidelity	0.15	3	5	1	5
Low validation burden	0.15	4	1	4	1
Instructor usefulness	0.10	4	3	3	3
Future extensibility	0.10	3	4	2	4
Cost and equipment requirements	0.05	5	3	5	2
Weighted total	1.00	4.35	2.40	3.15	2.60

The training path scores highest (4.35/5.00), driven by buildability, measurable evidence, and low validation burden.

7.8 Requirement Traceability Matrix

Table 16: Requirement traceability matrix

Req.	Stakeholder need	Design feature	Evidence source	Validation method	Status
FR-1 Load local module	Educators need updateable local tray rules.	JSON tray module and module selector.	Count-sheet and local tray evidence [?].	Change module data and reload.	Ready for impl.
FR-2 Retrieval cards	Learners need active recall of names, features, and counts.	Prompt-before-reveal cards.	Retrieval and flashcard evidence [?]; [?].	Card walkthrough and quiz logs.	Ready for impl.
FR-3 Pre/post assessment	Reviewers need measurable learning evidence.	No-hints pre-test and post-test variants.	Ofstead pre/post pattern [?].	Paired run export.	Protocol-ready.
FR-4 Quiz mode	Learners need feedback before applied sorting.	Quiz with confidence and weak-item logging.	Feedback and retrieval evidence [?].	Quiz result export.	Ready for impl.
FR-5 Practice feedback	Learners need actionable correction.	Error-specific feedback engine.	Tray defect categories [?]; [?].	Seeded error cases.	Defined.
FR-6 Metrics export	Instructors need reviewable evidence.	Attempt and item CSV/JSON.	Error-reporting and competency evidence [?]; [?].	Schema validation.	Defined.
FR-7 Mode separation	Assessment must remain fair.	Mode state and feedback suppression.	Assessment design and answer-leakage risk.	UI walkthrough.	Defined.
NFR-1 Fast data load	Demo cannot stall.	Local file storage.	Usability expectations.	Load timing <= 2 seconds.	Future test.
NFR-2 Short flow	Participants can finish in one session.	Bounded module and timed segments.	Study protocol and sample feasibility.	Timing <= 20 min for demo.	Future test.
NFR-3 Export reliability	No completed run should be lost.	Local export after each run.	Study evidence requirement.	100% completed-run exports.	Future test.
NFR-4 Privacy	Learners should not be surveilled unnecessarily.	Hashed IDs and no PHI.	Healthcare data governance concerns.	Export review.	Defined.
NFR-5 Content traceability	Local content must be auditable.	Source notes and instructor verification status.	Count-sheet and local content boundary.	Module review.	Defined.
NFR-6 Capability labels	Stakeholders must not overread scores.	Report language: simulated practice evidence.	Certification and clinical-transfer boundaries.	Report review.	Defined.
NFR-7 Accessibility	The interface must be usable by learners with varying abilities.	Audio controls, status messages, and color-independent indicators follow WCAG guidelines [?, ?, ?].	Accessibility audit and user testing.	Design review.	Future test.

8 Final Product Concept And Design Rationale (Matthew, Andy, Qiyu)

8.1 Final Product Concept

TrayGuard is a local tray training and assessment platform. The core product is a learner-facing app plus a file-backed tray module that defines the instruments, aliases, required counts, distractors, lookalike pairs, distinguishing features, and assessment variants for one local tray. The first build should use printable AprilTag instrument cards as the physical simulation layer. Each card represents one instrument or instrument variant. The app scans the cards on a tabletop or tray area, treats the detected markers as the learner's tray selection, scores the result against the tray template, and returns missing, extra, wrong, misidentified, and wrong-count feedback in practice mode. Pre-test and post-test modes suppress hints and corrective feedback so the same system can collect baseline and post-training evidence. The full learning loop is:

```
local tray module -> pre-test -> retrieval-first cards -> quiz -> AprilTag-card tray sort -> post-test ->
    metrics export
```

This concept keeps the prototype buildable while preserving the task structure: the learner must use local names, local counts, visual/semantic distinctions, and tray rules to assemble a simulated tray.

8.2 Alternatives Considered

Several product directions were considered:

Table 17: Alternative product directions considered and rejected

Alternative	Benefit	Reason rejected or deferred
Deployment-grade CV tray checker on real instruments	Closest to the original automation concept and strongest if validated in an SPD.	Requires local instrument data, clinical workflow integration, failure-mode validation, and regulatory/adoption work beyond our project timeline.
Software-only study and quiz module	Fastest implementation path and easiest baseline study.	Too far from the tray reconstruction task; it would mostly test recognition, not applied count-sheet use.
Real teaching instruments with camera-supported tray review	Better physical fidelity than cards.	Requires access to representative instruments, cleaning/handling rules, storage, and educator review; appropriate for an SPD pilot after the simulated loop works.
Printable QR/AprilTag instrument cards	Cheap, portable, repeatable, and compatible with device scanning; closely matches NeuroVase-style tangible cue-card learning.	Less physically realistic than real instruments, so claims must stay limited to simulated tray-task improvement.
Manual drag-and-drop tray simulation	Simple software implementation without camera setup.	Does not exercise physical search, placement, or scanning workflow, and is less persuasive for an interactive project demo.

The selected first prototype is the printable AprilTag-card path because it creates a physical, measurable simulation without depending on deployment-grade instrument recognition.

8.3 Selected Candidate and Backup Paths

The selected candidate is:

> A local tray training and assessment app that uses printable AprilTag instrument cards for simulated tray sorting, retrieval-first study, quiz, practice feedback, no-hints pre/post assessment, and metrics export.

The backup paths are layered so the project still produces evidence if one piece fails.

Table 18: Backup path strategy by risk level

Level	Selected path	Backup path	Claim preserved
Learning feature	Retrieval cards, quiz, practice sort, pre/post sort.	If the full loop is too long, run pre-test, study cards, practice sort, and post-test only.	Immediate simulated learning.
Physical interface	AprilTag cards scanned on a tray mat.	Use QR codes or manual card selection if AprilTag scanning is unreliable.	Local tray-task learning, with marker reliability reported as a limitation.
Content	`basic_general_tray_v1` with 8-12 required concepts, distractors, lookalikes, and matched variants.	Reduce to 6-8 required concepts while preserving at least two lookalike pairs and count errors.	Bounded module feasibility.
CV	Marker detection only for project scoring.	Defer all camera recognition and score user selections manually.	Training evidence without CV overclaim.
Evaluation	8-12 baseline participants plus delayed retention if schedule allows.	Run a smaller formative walkthrough and report it as usability/protocol evidence, not learning proof.	Risk-reduction handoff.
Content validity	SPD educator or instructor review.	Faculty or surgical-technology reviewer plus explicit future SPD review requirement.	Plausibility review, not clinical validation.

This backup strategy is intentionally conservative. It protects the project from collapsing into a fragile detector demo and keeps the main deliverable aligned with the training pivot.

8.4 Rationale for AprilTag Marker Choice

The prototype uses printable AprilTag fiducial markers on physical instrument cards. This section justifies AprilTag specifically against the alternatives of no physical component, QR codes, ArUco markers, and real instruments with camera-based recognition.

8.4.1 Why Physical Markers at All

A purely digital drag-and-drop tray simulation would be simpler to implement and would avoid camera-setup and detection-reliability concerns. It is rejected because the target task is fundamentally physical: SPD technicians search for, pick up, examine, count, and arrange real instruments under a tray boundary [?]; [?]. A purely on-screen interface removes the spatial search, the need to distinguish similar objects in hand, and the physical count-and-place workflow that the study is meant to simulate. Physical cards preserve those cognitive demands while replacing regulated stainless-steel instruments with safe, cheap, tractable tokens.

8.4.2 Why AprilTag over QR Codes, ArUco, or Real Instruments

AprilTag was chosen over QR codes, ArUco markers, and real instruments for reasons of detection robustness, error correction, and study feasibility. Compared with QR codes, AprilTag provides faster corner detection at sub-pixel precision, a lexicode-based Hamming-distance guarantee that corrects bit errors from glare or occlusion, and native pose estimation for future spatial feedback [?]; [?]. Compared with ArUco, AprilTag's lexicode dictionary guarantees a larger minimum inter-marker distance, and its corner-refinement pipeline produces more stable localization across lighting and image scales [?]; [?]; [?]. Compared with real instruments — the closest alternative to the original CV-checker goal — physical cards avoid the regulatory, handling, and recognition burdens that would dominate the engineering schedule, and follow the same abstraction strategy as NeuroVase's tangible cue cards [?]; [?]. The recommended tag family is `tagStandard52h13` (52 data bits, minimum Hamming distance 13), supported through OpenCV's `aruco` module with no additional library dependency [?]; [?].

8.5 Rationale for Gamified and Simulation-Based Learning Style

TrayGuard embeds retrieval practice, feedback, scoring, progressive difficulty, and confidence capture within a structured learning session. Each design element maps to an established mechanism (summarized in Section 6): pre-testing improves later retention, retrieval cards exercise active recall, weak-item logging enables targeted remediation, error-category feedback supports deliberate practice, and parallel no-hints post-tests prevent score inflation from layout memorization [?]; [?]; [?]; [?]; [?].

The design deliberately avoids leaderboards, badges, and competitive ranking, which can shift motivation from intrinsic learning to extrinsic reward-seeking [?]. Full VR simulation (e.g., SteriBoost) and narrative-driven modules (e.g., Touch Surgery) were considered but rejected because they add hardware or authoring burden without increasing the number of tray-repetition cycles per session [?]; [?]. The chosen level is closest to the serious-game approach of SteriDefi—a question-bank and scoring system that motivates repeated knowledge retrieval [?].

8.6 Training Session Walkthrough

Whereas the transparent box model (Section 7.5) decomposed the training flow into technical subfunctions, the following walkthrough describes the same session from the learner's perspective.

The learner sits at a table with a printed count sheet, a pool of AprilTag instrument cards face up, and a camera-connected device running the TrayGuard app. The session proceeds through eight steps.

Intake. The app displays a consent form and a brief background questionnaire (prior SPD experience, certification status, familiarity with surgical instrument trays). After submission, the app seeds a random assessment variant and displays a prompt: "Assemble the tray shown on the count sheet using the cards in front of you."

Pre-test. The app enters assessment mode: no hints, no corrective feedback, no timer visible to the learner but duration logged internally. The learner places cards one at a time on a marked tray area. The camera detects each card's AprilTag, maps it to an instrument ID, and adds it to the running selection. The learner presses Confirm when finished. The scoring engine classifies each required item as correct, missing, extra, wrong, or misidentified, and each correct item as correct-count or wrong-count. The accuracy score, error breakdown, and per-item confidence are recorded but not shown.

Retrieval-first study cards. The app switches to study mode. For each instrument in the tray module, it shows a prompt card: the instrument name, image, aliases, distinguishing features, and required quantity. The learner taps or clicks to reveal the answer side with full details. The app records which prompts the learner could not answer before revealing.

Quiz. The app presents identification questions drawn from the same instrument set. Each question shows an instrument image or name and asks for the corresponding name, aliases, features, or count. After answering, the learner rates confidence (1–5 or `Not sure`). The app records correctness, confidence, and response time per item and flags weak items (wrong or low-confidence answers).

Practice tray sort. The learner repeats the same physical card-sorting task as the pre-test. This time, the app provides immediate feedback: after each card is placed, it indicates whether the item is correct, wrong (wrong instrument placed), or extra (not on the tray list). After Confirm, the app shows the full error-category breakdown with correct, missing, extra, wrong, misidentified, and wrong-count counts, and highlights each error in the tray visualization.

Weak-item review. If the study protocol includes this step, the app redisplays retrieval cards for items the learner missed or answered with low confidence during the quiz or practice sort, allowing focused re-study before the post-test.

Post-test. The app switches back to assessment mode with a parallel tray variant (different instrument subset, same lookalike pairs, matched difficulty). The procedure matches the pre-test: no hints, no feedback, timed but no visible timer. The learner sorts cards and confirms. The scoring engine records the same metrics for pre/post comparison.

Summary and export. The app displays a session summary: pre-test and post-test accuracy scores, error-category counts side by side, confidence calibration (high-confidence errors, low-confidence correct answers), and a list of weak items that did not improve. The instructor or facilitator receives a CSV/JSON export with attempt-level and item-level records, hashed learner ID, module version, and assessment variant IDs.

9 System Design (Matthew, Jacob)

9.1 System Overview

TrayGuard should be built as a local module runner plus a card-scanning assessment surface. The app does not need to recognize real instruments for the first study. It needs to observe which simulated instrument cards the learner placed in the tray area, score that selection against a local tray template, and export evidence.

```
local tray module
  -> pre-test AprilTag tray sort
  -> retrieval-first study cards
  -> quiz with confidence / Not sure
  -> practice AprilTag tray sort with feedback
  -> post-test AprilTag tray sort
  -> metrics export
   -> committed 7-day delayed retention sort (see falsification criteria in Section~\ref{sec:study-2})
```

The architecture follows the evidence base:

- Local tray modules are required because count sheets define tray names, contents, quantities, sizes, and reference/catalog numbers, and real count sheets include fields such as reference number, description, location, quantity, check box, total count, and hospital signature [?]; [?]; [?].
- Retrieval-first study and quiz modes are justified by retrieval practice, health-professions flashcard evidence, and spaced digital education [?]; [?]; [?].

- Pre/post practice, confidence capture, and optional delayed retention follow the sterile-processing precedent in Ofstead and the retention warning in Bell et al. [?]; [?].
- AprilTag cards are a physical proxy for instrument selection and counting decisions. The tangible-card idea is supported by NeuroVase-style cue cards and board/card-based healthcare education, but clinical transfer remains future validation [?]; [?].

9.2 Subsystem Breakdown

The system separates into these functional subsystems:

Table 19: Functional subsystem breakdown

Subsystem	Responsibility
Local content/tray module	Stores instrument records, aliases, photos, distinguishing features, required quantities, distractors, and tray-template versions.
Learning workflow	Routes the learner through pre-test, study, quiz, practice sort, weak-item review, post-test, and export.
Retrieval practice	Presents prompt-before-answer cards and quizzes for instrument names, features, lookalikes, and required counts.
AprilTag card scanning	Detects printed marker cards and maps marker IDs to instrument IDs so the app can observe a physical tray simulation.
Simulated tray scoring	Compares detected cards and selected quantities against the tray template.
Feedback engine	Reports missing, extra, wrong, misidentified, and wrong-count errors in practice mode.
Confidence/uncertainty capture	Records confidence or `Not sure` responses for assessment and instructor review.
Reporting/export	Produces pre/post accuracy, time, error-category, confidence, and weak-item summaries.
Instructor review	Lets an educator verify tray content, local naming, distractors, and study evidence before broader use.

Computer vision for recognizing real instruments remains a future support subsystem. In the project build, the marker-scanning subsystem is a proxy for observing user decisions, not an instrument-identification system.

Figure 5: Functional block diagram showing data flow among subsystems. Rectangles represent the nine functional subsystems from Table 19; arrows indicate the primary direction of data or control flow.

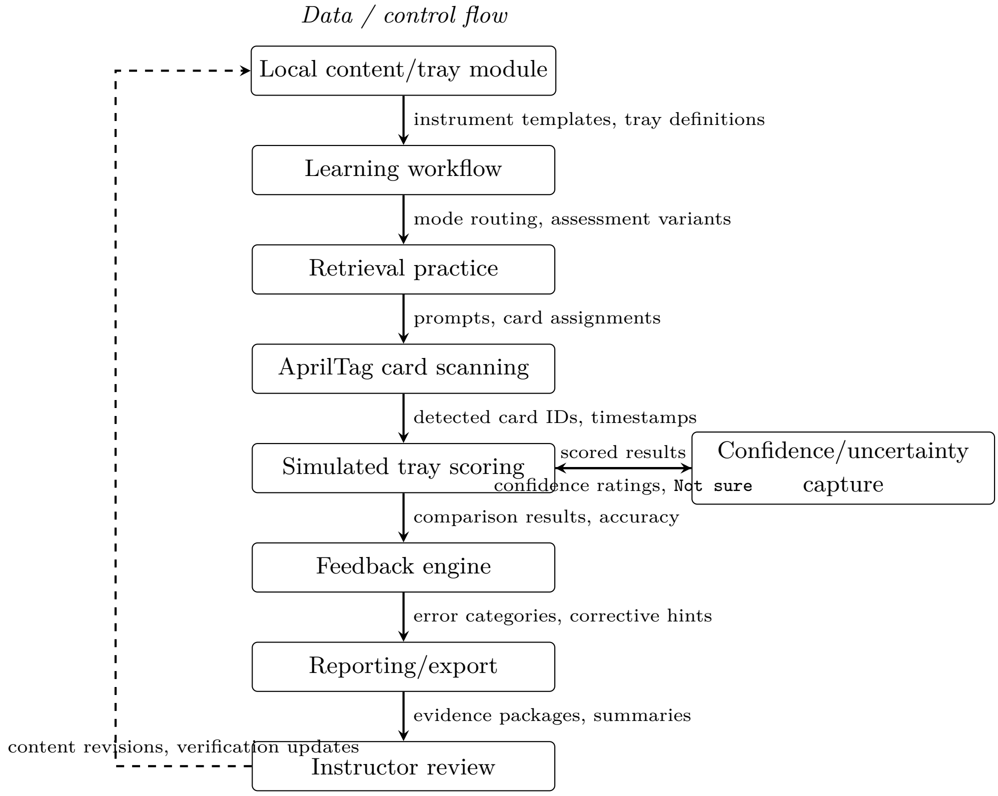
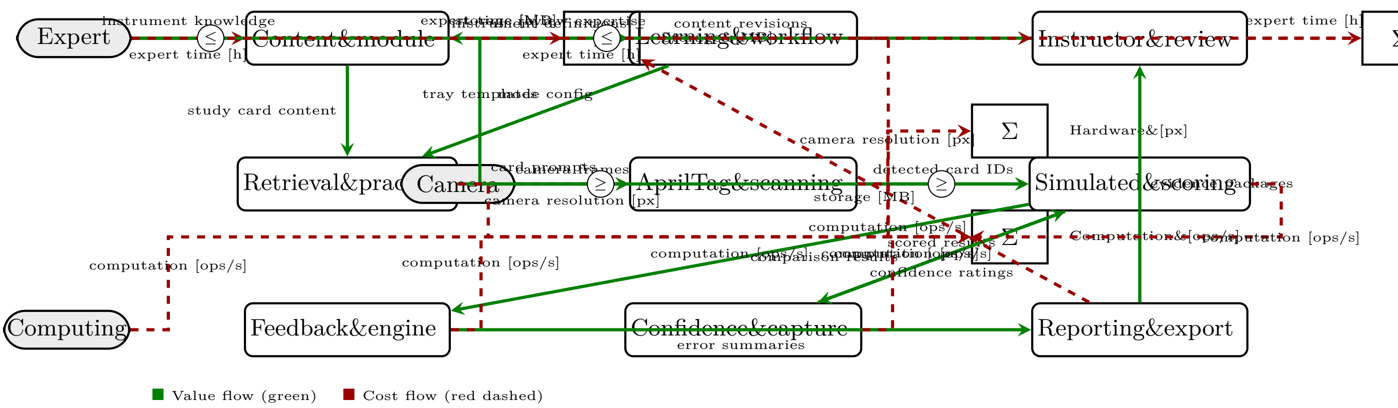


Figure 6: Dependency diagram showing cost (red dashed) and value (green solid) inputs/outputs for the nine subsystems. External inputs (left) feed into the system; value flows between blocks (green); cost flows accumulate into right-side summation blocks (red). Constraint icons (≤ / ≥) mark the highest-risk dependency wires.



9.3 Data Model and Module Contract

The data model should mirror a real count sheet while staying small enough for the first build. Stryker's public count sheets show the practical minimum: reference number, description, location/layer, quantity, check box, total count, additional items, and hospital signature. TrayGuard adds learning-specific fields for aliases, lookalikes, confidence, and error categories.

Table 20: Data model objects and required fields

Object	Required Fields	Why It Matters
`Instrument`	`id`, `display_name`, `family`, `aliases`, `distinguishing_features`, `image_refs`, `apriltag_id`, `local_verification_status`	Supports retrieval cards, quiz prompts, card scanning, and local naming.
`TrayTemplate`	`id`, `name`, `version`, `procedure_family`, `source_note`, `required_items`, `distractor_items`, `lookalike_pairs`, `assessment_variants`	Stores the local count-sheet task and separates required instruments from distractors.
`TrayItemRequirement`	`instrument_id`, `quantity`, `reference_number`, `location_or_layer`, `acceptable_substitutes`, `notes`	Mirrors real count-sheet fields and makes count/location errors measurable.
`AssessmentVariant`	`id`, `mode`, `random_seed`, `required_item_ids`, `distractor_item_ids`, `card_front_policy`, `feedback_enabled`	Keeps pre/post/delayed tasks comparable while preventing answer leakage.
`LearningRun`	`run_id`, `learner_id_hash`, `participant_group`, `tray_template_id`, `variant_id`, `mode`, `started_at`, `completed_at`	Separates study, quiz, practice, pre-test, post-test, and delayed attempts.
`DetectedCard`	`run_id`, `apriltag_id`, `instrument_id`, `timestamp`, `detection_confidence`, `confirmed_by_user`	Lets marker failures be audited instead of silently counted as learner errors.
`AttemptResult`	`run_id`, `selected_items`, `duration_seconds`, `accuracy_score`, `required_recall`, `errors`, `confidence_summary`	Provides the main pre/post evidence.
`ItemResult`	`run_id`, `instrument_id`, `required_qty`, `selected_qty`, `error_category`, `confidence`, `not_sure`	Supports item-level feedback, weak-item review, high-confidence error reporting, and instructor review.

This format follows the same trajectory as annotation standards in computer vision. The Pascal VOC dataset was introduced in 2005 by a university research group as a standardized benchmark for object detection [?]. It defined a common annotation format — bounding boxes, class labels, train/val/test splits — that allowed researchers to compare methods on a shared task. That format was not an official standard; it gained acceptance because the annotation schema mapped cleanly to the detection problem, the evaluation protocol was consistent, and the tooling (evaluation scripts, visualization code) made it usable across research groups. COCO followed the same model in 2014, extending the format to segmentation, captioning, and keypoint detection with a richer annotation structure and a public evaluation server [?]. Both formats succeeded because they captured correct domain semantics, enforced structural constraints, and came with tools that validated and consumed the format — not because a standards body approved them.

The tray module schema follows the same pattern. Required-item references are validated against instrument definitions. Lookalike pairs must resolve to valid instrument IDs. Assessment variants must reference existing items, and the scoring engine enforces these constraints at load time. The schema maps real count-sheet fields (reference number, description, location, quantity) while adding learning-specific fields for aliases, lookalike pairs, and confidence capture. The specific instrument data remains provisional — every instrument carries `local_verification_status: "pending_review"` — but the format itself is a well-specified hypothesis about how to represent local tray knowledge. Expert review (Study 3 in Section 11) tests that hypothesis, the same way VOC and COCO validation studies tested whether their annotation schemes captured the task correctly. If Study 3 identifies structural gaps — missing fields, incorrect relationships, wrong constraints — the schema is revised. That is how formats gain credibility: through iterative validation against the domain, not through the creator's institutional credentials.

The first implementation can store these objects in JSON files and export CSV for analysis. The constraint is schema stability: pre-test, post-test, and delayed retention must produce comparable rows. Sample instrument JSON:


```
{
  "id": "fgvc12_major_scissors_mayo_straight_7in",
  "display_name": "Scissors Mayo Straight 7\"",
  "family": "Cutting/dissecting",
  "aliases": [
    "SCISSORS MAYO STRAIGHT 7\"",
    "JARIT_101-220"
  ],
  "distinguishing_features": [
    "Heavy straight blades, semi-blunt tips",
    "Heavier, broader blade profile than Metzenbaum scissors",
    "Ring-handled cutting instrument"
  ],
  "common_confusions": [
    "fgvc12_major_scissors_mayo_curved_7in",
    "fgvc12_major_scissors_metz_curved_7in"
  ],
  "image_refs": [
    {
      "id": "view_a",
      "path": "data/instruments/fgvc12_major_tray_v1/fgvc12_major_scissors_mayo_straight_7in_view_a.jpg",
      "source": "fgvc12_google_drive_major_tray",
      "approved_for_study": true,
      "approved_for_assessment": false,
      "attribution": "FGVC12 shared Google Drive major tray image"
    },
    {
      "id": "view_b",
      "path": "data/instruments/fgvc12_major_tray_v1/fgvc12_major_scissors_mayo_straight_7in_view_b.jpg",
      "source": "fgvc12_google_drive_major_tray",
      "approved_for_study": true,
      "approved_for_assessment": false,
      "attribution": "FGVC12 shared Google Drive major tray image"
    }
  ],
  "apriltag_id": 0,
  "card_front_policy": "neutral_assessment_front",
  "study_prompts": [
    {
      "id": "name",
      "prompt": "Name this major-tray instrument.",
      "answer": "Scissors Mayo Straight 7\""
    },
    {
```

```
      "id": "distinguishing_feature",
      "prompt": "What feature distinguishes this from Metzenbaum scissors?",
      "answer": "Heavier, broader blade profile."
    }
  ],
  "notes": "FGVC12 major tray class JARIT_101-220. Expert review pending before assessment use.",
  "local_verification_status": "pending_review"
}
```

9.4 Mode Behavior

Table 21: Training mode behavior by hints, feedback, and metrics

Mode	Hints	Corrective Feedback	Metrics	Use
Pre-test	No	No	Yes	Baseline simulated tray-task proficiency.
Study cards	Yes	Yes	Optional	Prompt-before-reveal retrieval of names, aliases, families, features, and counts.
Quiz	Limited after answer	Yes	Yes	Retrieval practice, confidence calibration, and weak-item discovery.
Practice sort	Yes	Yes	Yes	Applied retrieval and deliberate practice on tray organization.
Post-test	No	No	Yes	Comparable same-day assessment after practice.
Delayed retention	No	No	Yes	Checks whether gains persist after 7 days (with falsification criteria).

9.5 Learning Loop

The learning loop (Section 8) applies retrieval practice as the instructional engine. The learner should usually commit to an answer before the system reveals it: name the instrument, identify a distinguishing feature, choose the lookalike, recall the count, or decide whether the item belongs in the tray. The simulated tray sort is applied retrieval: the learner must use those same facts in context with distractors and quantities.

Each feature in the learning loop maps to a specific evidence-based role:

Table 22: Feature-to-evidence mapping for the learning loop

Feature	Evidence-based role	Engineering requirement
Retrieval-first study cards	Initial instruction that still asks the learner to answer before reveal	Store photos, names, aliases, functions, distinguishing features, and local notes; present prompts before explanations
Identification quiz	Central retrieval practice for instrument identity and features	Ask recall or recognition questions, randomize distractors, record correctness, confidence, and time
Simulated tray sorting	Safe applied retrieval in a realistic work task	Let learners assemble or check a tray against a count sheet in a low-risk training environment
Immediate error-specific feedback	Deliberate practice and error correction	Explain wrong answers and classify errors as missing, extra, wrong, misidentified, or wrong count
Confidence ratings	Metacognitive signal for learner and instructor	Capture confidence before or after tasks so low-confidence correct answers and high-confidence errors are visible
Separated pre/post tests	Evaluation not contaminated by coaching	Disable hints and feedback during tests; export comparable pre/post accuracy, time, error type, and confidence

9.6 Module Scope

The first module should contain:

- 8-12 required instrument concepts;
- 3-6 distractor instruments;
- 2-4 known lookalike or same-family pairs;
- local names, aliases, family labels, required counts, and distinguishing features;
- AprilTag cards that represent instruments during the physical tray simulation;
- parallel pre-test, post-test, and delayed-retention tray variants with matched difficulty;
- instructor or expert review before results are presented as content-valid.

The cards should use neutral assessment fronts. Names, counts, and obvious answer labels should be hidden during pre-test, post-test, and retention tasks so the participant cannot solve the task by reading the card.

9.7 FGVC-Derived Module Plausibility

The FGVC-derived modules are practice modules built from the major-tray image classes reported by Atabuzzaman et al. [**?**]. The source contributes instrument names and images for a bounded image-card set: scissors, clamps, forceps, needle holders, retractors, suction, and other classes present in the major-tray archive. It gives TrayGuard a coherent pool of real instrument classes photographed under one dataset program.

Plausibility for these modules comes from three linked sources of evidence. First, every required card, distractor card, and lookalike pair is selected from the FGVC major-tray class mapping. Second, the module schema mirrors real tray/count-sheet fields: tray name, required item, quantity, reference number, location or layer, total count, and check status [?]; [?]. Third, the module size follows the project study constraint: 8-12 required concepts, 3-6 distractors, and 2-4 lookalike pairs, so novices can complete pre-test, practice, post-test, and delayed retention tasks in a single session. The FGVC-derived quantities are simulated as one card per selected class. The FGVC12 subset provides 16 instruments across four families (cutting/dissecting, clamping/occluding, grasping/holding, suturing), with six documented lookalike discriminations (Table 23). Each instrument in the tray module schema carries a `distinguishing_features` array and participates in `lookalike_pairs` that encode clinical feedback messages at the pair level:

- Mayo curved vs Metzenbaum curved: "Mayo scissors have heavier, broader blades; Metzenbaum are slimmer and lighter for delicate tissue."
- Kelly vs Crile: "Check serration pattern: Kelly serrations are distal half; Crile serrations run the full jaw."
- Allis vs Babcock: "Allis has multiple interlocking teeth; Babcock has smooth fenestrated jaws."
- Adson vs DeBakey: "Adson is short with 1×2 teeth; DeBakey is long with atraumatic longitudinal serrations."
- Kelly 6" vs 8": "Compare overall length: 6" vs 8"."
- Metzenbaum 7" vs 10": "Compare overall length: 7" vs 10"."

These feedback messages, the `distinguishing_features` arrays, and the lookalike pair structure are encoded in the module configuration (Section ??) and govern the feedback shown during quiz and sort modes.

Table 23: FGVC12 module instrument set, by family, with distinguishing features and lookalike relationships

Instrument	Distinguishing features	Lookalike(s)
Cutting / dissecting		
Mayo scissors straight 7”	Heavy straight blades, semi-blunt tips	Mayo curved 7”
Mayo scissors curved 7”	Heavy curved blades, broader than Metzenbaum	Metzenbaum curved 7”
Metzenbaum scissors curved 7”	Slim lightweight blades for delicate tissue	Mayo curved 7”, Metz curved 10”
Metzenbaum scissors curved 10” (distractor)	Same slim design, longer shaft	Metzenbaum curved 7”
Knife handle #3 short	Short handle, accepts #3 blade	—
Clamping / occluding		
Kelly clamp 6”	Serrations on distal half of jaw	Crile curved, Kelly 8”
Crile clamp curved 5.5”	Serrations run full jaw length	Kelly 6”
Kocher clamp 6”	Single tooth at tip, full serrations	—
Allis clamp 6”	Multiple interlocking teeth	Babcock 6”
Babcock clamp 6”	Smooth fenestrated jaws	Allis 6”
Kelly clamp 8” (distractor)	Distal-half serrations, longer overall	Kelly 6”
Babcock clamp 9.5” (distractor)	Larger smooth fenestrated jaws	—
Grasping / holding		
Adson forceps w/ teeth	Short, 1×2 teeth at tip	DeBakey 8”
DeBakey forceps 8” thoracic tip	Long, atraumatic longitudinal serrations	Adson w/ teeth
Suturing		
Mayo-Hegar needle holder 7”	Short heavy jaws, cross-serration	Mayo-Hegar 8”
Mayo-Hegar needle holder 8” (distractor)	Same, longer shaft	Mayo-Hegar 7”

9.8 Real Tray List Examples

The first TrayGuard module should be smaller than a clinical tray, but its data fields should look like real tray/count-sheet data. Three examples are useful:

Table 24: Real tray list examples and design implications

Real Example	What It Shows	Design Implication
Stryker Gamma4 Indication Tray count sheet	A real vendor count sheet uses tray name, insert/base sections, reference numbers, descriptions, locations, quantities, check boxes, total instrument counts, additional items, and hospital signature [?].	TrayGuard should store `reference_number`, `description/display_name`, `location_or_layer`, `quantity`, and `additional_items` fields even if the first module only uses names and quantities.
Stryker IMN Basic Instruments Tray count sheet	A second real count sheet uses the same pattern and splits the tray into insert/base groupings with total counts of 15 and 10 instruments [?].	The data model should support sections/layers and total-count validation, not just a flat list.
Multi-specialty tray rationalization project	Rubak et al. consolidated, separated, or restructured 1,340 different instrument trays across surgical specialties at one university hospital, finding that different specialties required different tray structures [?].	Real tray lists scale to thousands of configurations in a single hospital. The data model must be open-ended beyond the first study's 8–12 required concepts.
Published major orthopedic tray study	Toor et al. observed an 88-instrument major orthopedic tray across 80 procedures and compared optimized tray configurations of 47, 67, and 51 instruments [?].	Real clinical trays can be far larger than the first module. The first study should use 8-12 required concepts for feasibility, while preserving fields that can scale to larger trays.

For the first build, use a teaching-scale tray: 8-12 required concepts, 3-6 distractors, and 2-4 lookalike pairs. This is intentionally smaller than the real examples so a novice can complete the loop in 45-60 minutes.

10 Analysis (Matthew, Qiyu)

10.1 Prototype Capability Demonstration

The following capabilities have been demonstrated in the built prototype:

Table 25: Demonstrated prototype capabilities

Capability	Evidence	Status
Tray module loading	`basic_general_tray_v1` module loads and renders in the web app	Verified
Retrieval-first study cards	Prompt-before-reveal cards display instrument names, images, aliases, features, and tray quantities	Verified
Identification quiz	Quiz presents instrument images and captures name, confidence, and Not sure responses	Verified
AprilTag card detection	Camera-based detection maps marker IDs to instrument IDs in real time; manual entry fallback available	Verified
Simulated tray scoring	Submitted selections scored against tray template with accuracy and error categories	Verified
Error-category classification	Missing, extra, wrong, misidentified, and wrong-count errors computed per run	Verified
Pre/post assessment mode	No-hints mode with matched variants; baseline and post-training evidence collected	Verified
Confidence capture	1–5 scale and Not sure checkbox recorded per item; high-confidence errors flagged	Verified
CSV/JSON export	Attempt-level and item-level records exported with all required fields	Verified
Card generation	CLI generates printable marker PNGs, grid index sheets, and double-sided study flashcards	Verified

Every verified capability supports a direct link to the performance specifications (Section 7.3) and the requirements traceability matrix (Section 7).

10.2 Marker Reliability Analysis

The marker layer is a measurement tool: if detection fails, scores confound learner error with app error. The following analysis frames the reliability expectation:

Table 26: Marker reliability analysis

Dimension	Target	Method
Detection rate in normal layouts	$\geq$ 95%	Bench test with varied card spacing, rotation, and lighting
Detection rate in degraded conditions	Reported as limitation	Glare, partial occlusion, and extreme angles logged per test
Marker-ID confusion rate	0% in controlled test	All markers in the printed set decoded against ground truth
False positive rate (non-marker objects)	Reported as limitation	Objects placed in camera view that do not contain markers
User confirmation before scoring	Required	Confirmation screen shows detected cards for user review before scoring

The target values match the performance specification (Section 7.3) and the QFD requirement for marker reliability. Study 0 in the evaluation plan (Section 11.5) provides the experimental protocol for formal measurement.

10.3 Unit Test Results

Unit tests verify core logic that directly affects learner evidence quality:

- **Module loading.** JSON schema validation confirms that all required fields are present and correctly typed for the primary tray module.
- **Scoring accuracy.** Seeded tray submissions with known error patterns (one missing, one extra, one lookalike substitution, one wrong-count) return the expected category counts and overall accuracy score.
- **Export completeness.** CSV and JSON exports from scored runs contain all required fields at both the attempt level and item level.
- **Card generation.** Generated AprilTag PNGs decode to the correct marker IDs, and the double-sided flashcard layout renders instrument data without text overflow.

10.4 Design Traceability

Every design decision in the training path traces to an objective in the objective tree (Section 7.2), a specification in the performance specification analysis (Section 7.3), and a requirement in the requirement traceability matrix (Section 7).

11 Evaluation And Validation Plan (Matthew, Jacob)

11.1 Evaluation Framework

Two frameworks define the evaluation scope: Kirkpatrick's four-level model and Cook and Beckman's five-source validity framework [?, ?], which builds on the Standards for Educational and Psychological Testing [?]. A pre/post accuracy gain on a simulated tray task is Level 2 evidence. The current study ladder addresses Level 2 directly and builds toward Level 3 through convergent evidence and expert review, but does not claim Level 4 (reduced OR delays or hospital defect rates), which requires an SPD-site study [?].

Applied to TrayGuard, Cook and Beckman's validity framework requires the following evidence chain [?]:

1. **Content validity**: the tray module, names, aliases, counts, distractors, and lookalike pairs are plausible for a training tray. Addressed by expert review (Study 3) and instructor-verified module content.

2. **Response process**: learners interact with the system as intended; confidence ratings and `Not sure` capture reflect genuine self-assessment rather than confusion or guessing. Addressed by usability observation and confidence-calibration analysis (Studies 1-2).

3. **Internal structure**: error categories behave consistently: lookalike pairs produce misidentified errors, distractors produce extra errors, and repeated practice reduces specific error types. Addressed by pre/post error-type analysis (Studies 1-2).

4. **Relations to other variables**: TrayGuard tray-sort scores correlate with an independent measure of the same knowledge. This is the weakest link in the current design. Addressed by a convergent photo-identification test added to Study 1.

5. **Consequences**: the scores support the intended interpretation (improved simulated tray familiarity) and do not support unintended interpretations (clinical competence, SPD readiness). Addressed by scope boundaries and report labeling (Section 12).

A sixth issue is **population generalizability**. Novices start near zero accuracy, while SPD workers may already be proficient on familiar trays. Three responses apply: (i) Ofstead et al. show experienced SP workers are not at ceiling on novel tasks (mean pre-test 41% on borescope inspection) [?], so TrayGuard's value for workers is error-type reduction, confidence calibration, and retention maintenance rather than absolute accuracy gain; (ii) the dependent variable shifts from pre/post accuracy gain (baseline) to error-type reduction, confidence calibration, and new-module learning rate (workers), as Study 4 must specify [?]; and (iii) the learning mechanism (retrieval practice, feedback, spaced repetition) is population-independent even if the effect size is not [?, ?, ?, ?]. The study ladder (Table 27) maps each study level to the evidence source it earns; no single study provides all sources, and the claim is supported by the accumulated pattern.

11.2 Current Evidence Available

The current evidence package includes:

- literature-backed problem definition for SPD tray reconstruction and assembly errors;
- support for simulation, retrieval practice, flashcards, feedback, tangible card interfaces, and board/card-based healthcare education;
- a bounded final product concept based on local tray modules and AprilTag instrument cards, with the evaluation framework above clarifying which claims each evidence source supports;
- a subsystem architecture, data contract, mode behavior, and scoring rubric;
- a user-study plan that addresses content validity, response process, internal structure, relations to other variables through convergent evidence, and retention through a committed delayed check.

11.3 Conservative Baseline Study Plan

The target population for TrayGuard is SPD technicians who hold certification but need local tray familiarity. The first feasible study uses a student convenience sample as a conservative baseline: if the measurement instrument cannot detect learning in novices, it will not work with certified technicians. The study should evaluate whether the AprilTag-card simulation teaches a bounded local tray module, not whether participants are ready for real SPD work. Use one tray module with 8-12 required instruments, a distractor pool, and parallel pre/post tray variants. A reasonable first sample is 8-12 participants for classroom evidence; a larger follow-up can use paired statistics once the prototype and protocol are stable. Study sequence:

1. Consent and prior-experience questionnaire.
2. No-hints pre-test: assemble the simulated tray from AprilTag instrument cards using a tray list or count-sheet-style prompt.
3. Retrieval-first study cards for instrument names, aliases, distinguishing features, and counts.
4. Short quiz with confidence or `Not sure` capture.
5. Practice tray sort with immediate error-specific feedback.
6. Weak-item review for missed or uncertain instruments if time permits.
7. No-hints post-test on a comparable tray variant.
8. Short interview about confusing instruments, feedback usefulness, and perceived realism.

Primary measures are pre/post tray accuracy, duration, missing/wrong/extra/ misidentified/wrong-count errors, confidence calibration, high-confidence errors, low-confidence correct answers, and weak-item recovery. The study supports the training claim if most participants improve accuracy without a severe-error increase, maintain or improve completion time, and can explain at least one corrected error after practice.

The primary value of the baseline study is **instrument validation**, not population-effect estimation. The study answers five prerequisite questions that apply to any subsequent population:

1. Do the error categories behave as expected (lookalike confusion vs. missing vs. extra items), or does the scoring rubric produce noise?
2. Does the convergent photo-identification test show a moderate-to-strong correlation with tray-sort accuracy ($r \geq 0.50$)? If not, the tray scores do not measure instrument knowledge in *any* population.
3. Is confidence calibration measurable (high-confidence error rate, `Not sure` frequency), or do participants all guess at the same rate?
4. Does usability friction (scan failures, card confusion, UI navigation) dominate the learning signal, or can participants interact with the system as intended?
5. Which distractors are effective, which feedback messages help, and which instruments are confusing? These content-refinement findings are transferable to an SPD module.

If these fail in novices, they fail everywhere, because they are properties of the measurement instrument, not of the participant population. An SPD pilot run before the measurement tool is validated produces ambiguous results: a null finding could mean the system does not help, or it could mean the scoring rubric, error categories, or convergent measure are inadequate. The baseline study decouples these two failure modes.

A corollary is that the baseline study does **not** establish the effect size for SPD technicians. The dependent variable shifts from absolute accuracy gain (baseline) to error-type reduction, confidence calibration, and retention maintenance (target population) (Section 11.4). The baseline results warrant a technician study by showing the instrument functions; they do not substitute for a technician study.

11.4 Study Ladder

Table 27: Study ladder by participant type and validity source

Study	Participants	Validity source	Kirkpatrick level
Study 0: marker reliability	1-2 project members, no learning claim	Measurement reliability (prerequisite)	–
Study 1: baseline simulated learning	8-12 college students (conservative proxy for measurement validation)	Response process, internal structure, relations to other variables (convergent)	Level 2 (Learning)
Study 2: delayed retention	Same Study 1 participants, 7 days later	Relations to other variables (temporal)	Level 2 (Learning, durability)
Study 3: expert review	1 SPD educator, instructor, or knowledgeable reviewer	Content	Level 2 (Learning, content plausibility)
Study 4: SPD technician pilot	Certified or in-training SPD technicians (target population)	Consequential, extrapolation to target context	Level 3 (Behavior)

The project should prioritize Studies 0-2 and, if possible, one expert review.

11.5 Study 0: Marker Reliability

Purpose: prevent app or camera failures from contaminating learner scores. Setup:

- Print all AprilTag cards used in the tray module.
- Use the same device, camera angle, lighting, tray boundary, and table setup planned for the user study.
- Test cards individually, in valid tray layouts, and in cluttered layouts with distractors.

Record total cards visible, cards detected, false positives, missed cards, duplicate or unstable detections, and lighting or occlusion notes. Acceptance target before user testing:

- at least 95% card detection in normal layouts;
- no systematic confusion between marker IDs;
- visible confirmation screen before final scoring so camera misses are not counted as learner errors.

11.6 Study 1: Baseline Simulated Learning

Recruit 8-12 college students or classmates with no formal SPD experience and no prior exposure to the module. Collect optional background variables such as medical, biology, lab, tool, or surgical-instrument familiarity. This follows the NeuroVase-style proxy choice: students are not the target population, but they are acceptable for a first conservative baseline study that validates the measurement instrument before deploying with SPD technicians. Use a within-subject pre/post design:

```
intake -> pre-test -> training -> practice -> immediate post-test -> survey/interview
```

Target one 60 minute session per participant:

Table 28: Study 1: baseline session timeline

Segment	Time	Activity
Consent and intake	3-5 min	Explain simulated study, collect background.
Device orientation	2 min	Show how to scan and submit without teaching tray answers.
Pre-test tray sort	5-8 min	No hints, no feedback, timed.
Retrieval cards	10-12 min	Prompt-before-reveal study of required items, distractors, features, and counts.
Quiz	5-8 min	Short recall/recognition quiz with confidence or `Not sure`.
Practice sort	8-10 min	Same style task with immediate feedback.
Immediate post-test	5-8 min	Parallel variant, no hints, no feedback, timed.
Convergent photo-ID test	5-7 min	Paper-based photo identification of each required instrument plus 2 unseen transfer-probe instruments.
Survey and interview	5-8 min	SUS or short 5-point ease/usefulness/confidence survey plus open questions.

Primary learning measures:

- tray accuracy: required items correctly selected with correct count;
- severe errors: missing, wrong, misidentified, extra, wrong-count;
- duration from task start to final submission;
- high-confidence errors;
- `Not sure` selections or low-confidence correct answers;
- weak-item recovery from quiz/practice to post-test.

Convergent evidence measures:

- **Photo-identification test**: after the post-test tray sort, learners complete a short paper-based test showing high-resolution photos of each required instrument and 2-3 distractor instruments. For each photo, they write the instrument name, indicate whether it belongs in the tray, and rate their confidence. This test shares the knowledge domain (instrument identity, aliases, lookalike discrimination) with the tray sort while differing in response format (identification without physical card selection). The target correlation between post-test tray accuracy and photo-ID accuracy is $r \geq 0.50$, following the convergent-validity standard that two measures of the same construct should show moderate-to-strong association [**?**, **?**, **?**].
- **Transfer probe**: the photo-ID test includes 2 instruments from a different tray module that the learner has never seen. If TrayGuard teaches general instrument-discrimination skill rather than module-specific memorization, learners who performed well on the trained tray should show better discrimination on unseen instruments. This is a weak test (novices have no basis to identify unseen instruments), but the pattern across participants is informative: even chance-level performance on

the transfer probe, when combined with strong within-module gains, supports the bounded claim of module-specific learning.

Usability and experience measures:

- completion rate;
- help requests;
- scan failures or rescan attempts;
- SUS or short 5-point ease/usefulness/confidence survey;
- open feedback on confusing cards, confusing instruments, and perceived realism.

This design follows precedents including NeuroVase (tangible AR medical learning with 40 participants, pre/post testing, SUS) [**?**], Tulipan et al. (Touch Surgery with novice medical students) [**?**], and Ofstead et al. (SP professional borescope training pilot) [**?**]. A future between-group design could isolate the incremental effect of simulated tray practice beyond flashcards alone, but the within-subject pre/post design provides simpler logistics and smaller sample requirements for the first prototype.

11.7 Study 2: Delayed Retention

Same-day post-testing supports immediate learning only. It should not be called retention [**?**]. The delayed check distinguishes immediate practice gain from retained learning. In Kane's validity argument framework, the extrapolation inference from test score to real-world competence depends on evidence that performance persists after a delay during which transient factors (memorization of exact layouts, test familiarity, short-term motivational effects) have decayed [**?**]. Without a delayed measure, the interpretive argument supports only a same-day practice-effect claim.

The primary retention interval is **7 days** from the initial session. This interval is a practical compromise: Bell et al. found that medical-knowledge gains from a single intervention could decay measurably within 7 days [**?**], and Larsen et al. showed that the testing-effect advantage over restudy persisted at least 7 days in a medical-education randomized trial [**?**]. Ofstead et al. used a longer (2-month) interval with a booster session and reported approximately 90% retention of the post-test gain at that follow-up, but their intervention included workplace homework between sessions [**?**]. For a single- session training tool without between-session homework, 7 days is the shortest credible retention interval that separates durable learning from ephemeral practice gain.

Procedure:

1. Ask whether the participant studied the material since the first session.
2. Run a no-hints delayed tray sort with a third parallel variant.
3. Capture confidence or `Not sure`.
4. Run the photo-identification test again to measure convergent-retention correlation.
5. Ask what they remembered and what decayed.

Falsification criteria. The retention claim is supported only if:

- delayed accuracy retains at least half of the immediate pre/post gain (i.e., delayed − pre $\geq 0.5 \times$ (post − pre));
- delayed accuracy does not return to within 5 percentage points of the pre-test baseline (i.e., the retention effect is not attributable to measurement noise or test-retest practice);
- at least 70% of Study 1 participants complete Study 2.

If these criteria are not met, the project claim reverts to immediate learning only and the retention gap is reported as a limitation that future work (addressable by distributed practice, multiple training sessions, or a booster) must close. This follows Kane's principle that the interpretive argument must include explicit rebuttals: the conditions under which the claim would not be supported [**?**].

Optional extended intervals. If the 7-day check succeeds and schedule permits, additional checks at 21-28 days or 2 months would strengthen the retention claim further. The 2-month interval would match the Ofstead booster precedent [**?**].

11.8 Study 3: Expert Review

Expert review addresses the main weakness of college-student testing: content validity. The reviewer can be an SPD educator, SPD supervisor, sterile-processing instructor, surgical technology instructor, or faculty member with relevant instrument knowledge. Materials to show:

- tray list;
- instrument cards;
- distractor list;
- aliases and distinguishing features;
- scoring rubric;
- sample learner report.

Questions:

1. Are the instrument names and aliases plausible?
2. Are the required counts and distractors plausible for a training tray?
3. Are the lookalike pairs educationally meaningful?
4. Are any feedback messages misleading or unsafe?
5. Would the weak-item report help target coaching?
6. What would need to change before using this with SPD trainees?

11.9 Analysis and Acceptance Bar

For 8-12 participants, report descriptive paired results:

- participant-level pre/post table;
- mean and median accuracy change;
- mean and median duration change;
- total error counts by type before and after;
- high-confidence errors before and after;
- weak-item examples;
- representative qualitative feedback.

Avoid strong inferential claims unless the sample is larger. If the sample is large enough, add paired differences with confidence intervals. The training claim is supported if most learners show:

- higher post-test accuracy;

- equal or lower post-test duration without accuracy decline;
- fewer severe errors, especially missing and misidentified items;
- fewer high-confidence errors;
- fewer `Not sure` responses or low-confidence correct answers;
- evidence that missed or uncertain items improve after repeated retrieval;
- interview evidence that they can name what they learned or what remains hard.

Best project claim wording:

> In a baseline college-student sample, TrayGuard improved immediate simulated tray-sorting performance after retrieval-centered practice. A delayed 7-day check tested short-term retention. A convergent photo-identification test provided evidence that the tray-sort scores reflect instrument knowledge rather than task-specific familiarity. Expert review was used to assess whether the tray content and feedback were plausible for future SPD trainee validation. These findings establish that the learning mechanism works in this task domain for a naive population. They do not establish the effect size for experienced SPD workers, whose baseline on familiar trays may be high and for whom the dependent variable shifts from accuracy gain to error-type reduction, confidence calibration, and retention maintenance (Section 11.4).

11.10 Validity Claims Summary

No single study provides all validity sources; the overall claim rests on the accumulated pattern [?, ?].

Table 29: Validity claims mapped to evidence sources

Claim	Validity source	Evidence	Would be falsified by
The tray module content is plausible for training.	Content	Study 3 expert review; instructor-verified module; literature grounding and real count-sheet fields (Section 9.8).	Expert finds names, counts, distractors, or feedback unrealistic; module uses content that contradicts published tray listings.
Learners interact with the system as intended; confidence reflects genuine self-assessment.	Response process	Study 1 usability observation; confidence-calibration analysis (high-confidence error rate, `Not sure` frequency); SUS scores; interview feedback about confusion points.	Frequent scan failures produce scores that reflect app error rather than learner knowledge; confidence ratings do not distinguish correct from incorrect items; SUS scores below 68.
Error categories behave consistently and decompose learning gains.	Internal structure	Study 1-2 pre/post error-type analysis; lookalike pairs produce proportionally more misidentified errors than non-lookalike items; practice reduces specific error categories rather than shifting errors across types.	Error types do not differ between lookalike and non-lookalike pairs; accuracy gain comes entirely from one error category (e.g., fewer missing items) while other categories remain unchanged.
Tray-sort scores measure instrument knowledge and not only task-specific familiarity.	Relations to other variables	Study 1 convergent photo-ID test; correlation between tray-sort accuracy and photo-ID accuracy; transfer probe on unseen instruments.	Photo-ID correlation $r < 0.30$; photo-ID scores show ceiling or floor effects that prevent discrimination; transfer-probe performance is negatively correlated with trained-tray performance.
Learning persists beyond the initial session.	Relations to other variables (temporal)	Study 2 delayed tray sort at 7 days; retention relative to pre-test baseline; comparison of delayed accuracy to immediate post-test accuracy.	Delayed accuracy returns to within 5 pp of pre-test; less than half of the immediate gain is retained; fewer than 70% of Study 1 participants return.
The report supports instructor coaching without supporting overclaim.	Consequences	Scope boundaries (Section 12); report labels frame results as simulated practice evidence; export fields include weak items and confidence calibration.	Report language uses pass/fail or clinical-readiness labels; results are used for employment decisions; learners report feeling surveilled rather than coached.
Learning gains in baseline study transfer to SPD workers (extrapolation).	Generalizability / extrapolation	Ofstead precedent: experienced SP workers were well below ceiling on novel tasks; mechanism (retrieval practice + feedback) is population-independent per the training literature; dependent	Worker study fails to show improvement on any worker-relevant outcome; baseline-effect-size extrapolation is cited without technician data.

11.11 Future SPD Workplace Pilot

A later SPD pilot requires local count sheets, local photos or real teaching instruments, SPD educator review, technician or trainee participants, approval/privacy review, and a comparison condition such as current study materials or clinical supervisor-led orientation. It should measure whether the system helps target users in a workplace-relevant training context. Only that later pilot can support claims about SPD trainee usefulness, supervised transfer, or operational training adoption.

The SPD technician pilot must change the dependent variable from the baseline study. Where baseline participants are measured on absolute accuracy gain, experienced workers are measured on: error-type reduction (fewer specific errors on known trays), speed improvement without accuracy loss, confidence calibration (high-confidence error rate), new-module learning rate (time to criterion on an unfamiliar tray), and retention maintenance (preventing decay on low-frequency trays). The pilot protocol must specify which of these applies and the relevant minimum viable improvement before data collection [**?**, **?**].

12 Risk Assessment And Scope Boundaries (Matthew, Jacob)

12.1 Formal Risk Register

Table 30: Formal risk register with causes, consequences, and mitigations

Risk	Cause	Consequence	Mitigation	Residual Boundary
Student proxy overclaim	Participants are not SPD workers.	Reviewers reject relevance to target users.	State that students test simulated learning as a baseline; add expert review; reserve SPD claims for future pilot.	No SPD worker effectiveness claim.
Same-day retention mistake	Immediate post-test follows practice.	Report overstates durability.	Add committed 7-day delayed check with falsification criteria; otherwise call it immediate learning only.	Retention claim only after delayed test.
Card physical realism	Cards do not reproduce weight, scale, tactile handling, or real visual lookalikes.	Transfer to real instruments remains unknown.	Use cards as selection/count proxy; add local photos or real teaching instruments later.	No clinical handling competency claim.
Marker detection errors	Lighting, occlusion, angle, or overlap causes missed detections.	App errors look like learner errors.	Run Study 0; log raw detections; show confirmation before scoring.	Marker reliability must be reported.
Answer leakage from cards	Labels or marker IDs reveal the answer.	Measured gain is not learning.	Use neutral assessment fronts; randomize marker IDs; hide answer text in assessment.	Study and assessment cards must be separated.
Content invalidity	Student-built module may use wrong names, counts, or distractors.	Results do not reflect plausible SPD training.	Require expert review of module content and feedback.	Content validity limited until SPD review.
Feedback contaminates assessment	Practice answers leak into post-test.	Pre/post comparison inflated.	Use parallel variants; suppress hints and feedback in assessment modes.	Cannot claim broad generalization from one tray.
Usability friction dominates learning	Scanning or workflow confusion slows participants.	Poor scores reflect interface friction.	Track help requests, scan failures, SUS/ease ratings, and qualitative feedback.	Usability results must be reported separately.

12.2 Unknowns and Concerns

The current design is viable as a low-cost formative learning prototype, but several concerns remain unresolved:

Table 31: Unknowns and concerns with strengthening actions

Concern	Why it matters	Strengthening action
Card-to-instrument transfer	AprilTag cards preserve selection and counting decisions, but they do not reproduce weight, scale, handling, tactile inspection, or real visual lookalikes.	Treat the first study as simulated learning only; add an SPD educator review and a later task using local photos or real teaching instruments.
Flashcard overreach	Flashcards support recall, but tray work requires applying local rules under distractor pressure.	Keep flashcards as preparation, then make the main outcome a no-hints tray sort with missing, extra, wrong, misidentified, and wrong-count scoring.
Marker identity leakage	If cards visibly encode names or IDs, learners may match labels instead of learning instruments.	Use neutral card fronts during assessment, randomize marker IDs, hide answer text, and separate study cards from assessment cards.
Content validity	A student-built tray module may contain wrong names, unrealistic distractors, or nonlocal quantities.	Require instructor or SPD educator verification of every instrument, alias, count, and distractor before using results as evidence.
ML regulatory compliance	Any future AI-powered feature must satisfy FDA Good Machine Learning Practice guidelines and transparency requirements [?, ?].	Treat CV features as future support layers and document the regulatory boundary between training support and clinical decision support.
Assessment equivalence	Pre/post gains are weak evidence if the post-test is easier or repeats the same exact layout.	Build parallel pre/post tray variants with matched item families, counts, distractors, and difficulty.
Short-term learning only	A same-day post-test can show practice effects rather than retention.	Add a delayed retention check or booster if schedule permits.
Baseline participant limits	Student participants can test learnability, but they do not represent SPD technicians under workflow pressure.	Report student results as baseline simulated-learning evidence and reserve adoption/clinical-transfer claims for an SPD technician pilot.
Scanning reliability	Camera angle, lighting, occlusion, and card overlap may produce app errors that look like learner errors.	Log raw detections, show a confirmation state before scoring, and run a marker-detection bench test before the user study.
Confidence calibration	Confidence ratings can become noise if captured too often or too vaguely; novice learners may also exhibit systematic overconfidence on items they cannot reliably identify [?].	Capture simple confidence or `Not sure` at assessment moments and analyze high-confidence errors separately.
Instructor usefulness	Better learner scores do not automatically mean the output helps educators.	Add a short instructor-facing report and ask an educator whether the weak-item summary would change coaching.

These holes define the claim boundary: the prototype can show whether a marker-card training loop improves performance on a simulated local tray task. It cannot yet show real SPD competence, reduced OR delays, or deployment-ready instrument recognition.

12.3 Scope Boundaries

Allowed after Study 1:

- the baseline population improved on a simulated tray task;
- the interface was usable for first-time learners;
- retrieval cards plus practice sorting produced measurable immediate gains.

Allowed after Study 2:

- gains persisted after 7 days (meeting the falsification criteria in Section 11.7);
- the system showed measurable short-term retention in a simulated tray task, distinguishable from same-day practice effects.

Allowed after Study 3:

- an expert reviewer found the module plausible for future SPD validation;
- content concerns were identified and revised before broader testing.

Not allowed yet:

- SPD technicians will improve;
- the system reduces tray errors in hospitals;
- the system reduces OR delays;
- the system certifies competency;
- the app recognizes real surgical instruments;
- same-day gains demonstrate retention.

13 Broader Impacts And Ethics (Matthew, Jacob)

TrayGuard's positive impact is a more accessible way to practice local tray knowledge before a learner consumes scarce supervised training time. If it works, learners get repeatable practice, instructors get clearer weak-item evidence, and future teams get a safer bridge between training content and camera-supported tray review. The ethical risk is that a useful training report could be misused. A learner score is not a certification result, employment screen, disciplinary record, or permission to perform unsupervised sterile-processing work. The report should therefore label outputs as simulated practice evidence and should avoid language such as pass, fail, competent, or cleared unless a formal program adds its own supervised competency process. Using the principles of beneficence, nonmaleficence, autonomy, and justice:

Table 32: Bioethics principles applied to TrayGuard

Principle	TrayGuard implication
Beneficence	The system should help learners improve and help educators target coaching, especially on missing, wrong, misidentified, and high-confidence errors.
Nonmaleficence	The system should not imply clinical readiness, replace hands-on supervision, or hide uncertainty behind a single score.
Autonomy	Learners and instructors should understand what is being recorded, why it is being recorded, and how the report will be used.
Justice	Access should not depend on having a perfect local instrument library, expensive hardware, or one preferred training site. Modules should be updateable and printable.

Negative externalities and controls:

Table 33: Negative externalities and ethical controls

Risk	Ethical concern	Control
Punitive learner metrics	Scores could become surveillance or discipline instead of coaching.	Use hashed IDs for studies; report weak items and practice progress; prohibit employment or competency decisions from prototype data.
Overreliance on simulated scores	A high score could be mistaken for real SPD readiness.	Label all results as simulated; require supervised hands-on validation for clinical claims.
Reduced supervised practice	Programs might use the tool to cut clinical supervisor time.	Frame TrayGuard as preparation for supervised practice, not replacement.
Unequal local content	Better-resourced sites may build richer modules.	Use open file formats, printable cards, and a minimal module template.
Instrument-library bias	Learners may only practice instruments represented in the module.	Version modules, list coverage gaps, and require local review before use.
Answer leakage	Labels or marker IDs could inflate learning results.	Use neutral assessment cards and hide names/counts in assessment modes.
Printing and material waste	Card iterations create paper/plastic waste.	Use reusable sleeves, print small modules, and revise digitally before reprinting.
Privacy creep	Future reports could collect names, staff IDs, or workplace behavior.	Keep patient data out of scope; collect only fields needed for learning analysis.

These controls are part of the design, not optional documentation. They protect the stakeholder value of TrayGuard by keeping the training tool honest about what it can and cannot prove.

14 Project Plan And Future Work (Matthew, Jacob)

14.1 Completed Work

The following subsystems have been implemented:

- **Training web app.** A FastAPI backend serves a single-page browser application with seven-step learner flow: intake, pre-test, study cards, quiz, practice sort, post-test, and summary. Camera capture through `getUserMedia` streams frames to a server-side ArUco detection endpoint. Manual quantity entry serves as a fallback when camera access is unavailable or unreliable `src/trayguard/web/app.py`.
- **Tray module data model.** Eight file-backed tray modules define required instruments, distractors, lookalike pairs, assessment variants, and ArUco marker mappings. The primary module, `basic_general_tray_v1`, contains 10 required instrument types (15 total required units), 5 distractors, 6 lookalike pairs, and 2 assessment variants with matched difficulty. Five instrument catalogs covering 400+ classes provide authoring reference material `config/tray_modules/`.
- **Card generation and printing.** A CLI command generates printable ArUco marker PNGs, a marker-grid index sheet for camera scanning, and double-sided study flashcards with instrument images, names, aliases, distinguishing features, and tray-specific quantities. An image-processing pipeline handles EXIF rotation, content-aware cropping, and automatic marker placement `src/trayguard/learning/cards.py`.
- **Scoring engine.** A scoring function classifies each item into correct, missing, extra, wrong, misidentified (lookalike substitution), or wrong-count categories. Error-point aggregation maps to an accuracy score and required-recall metric. Confidence ratings (1-5 scale and a `Not sure` checkbox) are captured per item and flagged as high-confidence errors or low-confidence correct answers `src/trayguard/learning/scoring.py`.
- **Export and run storage.** Each completed run produces a JSON payload plus CSV files at the attempt level and item level. Learner identifiers are hashed (SHA-256 truncated to 16 hex characters). The export schema includes all error categories, confidence metadata, and module version fields `src/trayguard/learning/export.py`; `src/trayguard/learning/run_store.py`.
- **Tests.** Unit tests verify module loading, scoring accuracy (full-credit and lookalike-substitution cases), CSV/JSON export file completeness, and ArUco marker generation and detection `tests/test_learning.py`.

14.2 Project Plan Postmortem

The training-centered pivot (Section 5) reduced risk by changing the main proof obligation from deployment-grade recognition to measurable learning in a baseline study.

14.3 Future Work Roadmap

Table 34: Future work roadmap phases

Phase	Goal	Exit Criteria
1. Minimal prototype	Build the end-to-end local tray loop.	Load module, scan cards, run pre-test, show study cards, run quiz, run practice sort, run post-test, export metrics.
2. Verified module	Make the first tray content defensible.	8-12 required instruments, distractors, lookalikes, neutral assessment cards, matched variants, and expert/instructor review.
3. Marker reliability	Prove the app can observe card choices.	At least 95% detection in normal layouts, no systematic ID confusion, confirmation screen before scoring.
4. Novice study	Test immediate simulated learning.	8-12 college-student participants, pre/post results, usability notes, and bounded claim language.
5. Retention check	Test whether learning persists.	7-day delayed no-hints tray sort with third matched variant, plus falsification criteria (Section 11.7).
6. Report refinement	Turn study output into project evidence.	Participant-level results, aggregate pre/post plots, error-category table, high-confidence errors, and qualitative themes.
7. SPD educator review	Improve content validity.	Reviewer feedback on names, counts, distractors, lookalikes, scoring, and weak-item report.
8. Future SPD pilot	Test target-user relevance.	SPD trainee or technician participants, local count sheet/photos, approval/privacy review, and comparison condition.
9. Optional CV support	Reintroduce real-instrument recognition as support, not the core proof.	Detector tested on local photos with generalization and failure-mode reporting.

14.4 GitLab Epic / Issue Map

Table 35: Epic and issue map for continuation

Epic	Issues
Training Module	Define first tray module; add instrument cards with aliases and features; add tray quantities and distractors; build study-card view; build identification quiz.
Scoring And Feedback	Record quiz correctness, duration, and confidence; build simulated tray sorting; implement missing/extra/wrong/misidentified/wrong-count scoring; add practice feedback.
Assessment	Build pre-test mode; build post-test mode; suppress hints and feedback during assessment; add delayed-retention mode if schedule allows.
Evidence And Reporting	Export learner metrics; generate pre/post summary; report confidence calibration, high-confidence errors, and weak-item recovery.
Pilot Evidence	Write pilot walkthrough script; run marker reliability check; run baseline study; record confusion points; note that the baseline uses non-SPD participants as a conservative proxy.
Content Validation	Prepare expert-review packet; collect reviewer feedback; revise names, counts, distractors, and feedback language.
Presentation	Lock final demo narrative around simulated learning, retention boundary, and future SPD validation.

14.5 Next Team Priorities

14.6 Budget

Table 36: Project budget by category

Category	Expected Cost	Notes
Software prototype	$0 direct cost	Uses existing local app stack and open-source AprilTag support where available.
Physical cards	$5-25	Printer paper/cardstock, sleeves, tape, or foam backing.
Tray surface	$0-30	Existing table, poster board, foam board, or simple marked tray boundary.
Camera/device	$0 if using existing laptop/tablet/phone	Optional stand improves reliability but is not required for the first study.
Participant cost	$0-100	Small snacks/gift cards if allowed; class demo can be unpaid.
Expert review	$0-200	Likely unpaid instructor feedback; paid SPD educator review would be stronger.
Future SPD pilot	TBD	Depends on site approval, staff time, privacy review, and local instrument access.

The main cost is labor: module authoring, prototype implementation, marker testing, study moderation, analysis, and report writing.

15 Discussion And Final Recommendations (Matthew, Jacob)

The next step is to build and test the software-first training loop before investing more effort in deployment-grade tray automation. The problem evidence justifies the target task, and the learning evidence justifies a simulated training prototype.

The evaluation plan targets Kirkpatrick Level 2 (Learning) with internal- structure, convergent, and temporal evidence; Level 3 and Level 4 remain future work [?]. The five-source validity framework provides the evidentiary structure with explicit falsification criteria [?].

Population generalizability is addressed by separating the dependent variable by population: baseline participants on absolute accuracy gain, workers on error-type reduction, confidence calibration, and new-module learning rate (Section 11.4). The Ofstead precedent confirms that even certified SP workers have room for improvement on novel tasks [?].

Stakeholders can conclude the following:

- Tray reconstruction targets readiness failures in assembly, visualization, identification, missing items, wrong items, extra items, and local count-sheet use.
- The training pivot avoids the validation burden of deployment-grade CV while addressing local tray knowledge and supervised practice bottlenecks.

- A marker-card prototype can support a bounded baseline study if the cards are neutral, detection is reliable, assessment variants are comparable, and results are labeled as simulated.
- The project defines the module contract, performance targets, error categories, study ladder, convergent validity target, and claim boundaries.
- The evaluation design addresses content validity (expert review), internal structure (error-type analysis), relations to other variables (convergent photo-ID test), and temporal persistence (committed 7-day retention check with falsification criteria).

Stakeholders cannot yet conclude that TrayGuard:

- certifies sterile-processing competence;
- replaces hands-on hours, clinical supervisor observation, or local sign-off;
- reduces OR delays or hospital tray-defect rates;
- works for SPD technicians in a real workplace;
- recognizes real instruments across manufacturers, lighting, occlusion, and tray layouts;
- produces the same effect size for experienced workers as for baseline participants (the dependent variable and expected magnitude differ by population);
- supports retention unless the 7-day delayed check meets the falsification criteria specified in Section 11.7.

The next team should therefore implement the learning loop described in Section 8, extended with a convergent photo-ID test. After that loop works, the team should run the marker reliability check, then a small baseline pre/post study with the convergent photo-ID measure, then the committed 7-day retention check, then an expert review of names, counts, distractors, lookalikes, and feedback. Only after those results should the project expand toward real teaching instruments, local photos, SPD trainees, or real-instrument CV. Final recommendation:

> Treat TrayGuard as a local tray learning and assessment platform first. Use the project prototype to validate the measurement instrument in a conservative baseline study, demonstrate that gains reflect instrument knowledge (convergent photo-ID evidence), persist for at least 7 days (retention check), and are grounded in plausible content (expert review). Those results justify an SPD technician pilot that measures error-type reduction, confidence calibration, and new-module learning rate in the target population.

16 References

A CV Experiment Methods and Additional Results

A.1 Brightness Experiment Setup

The brightness experiments used a self-collected dataset of five classes (four spoon variants and a fork) photographed on two backgrounds (matte white cardstock, reflective silver wrapping paper) across 11 brightness levels forming a descending ladder from 800 to 80 estimated lumens. Two layout conditions were captured: separated (instruments arranged in two distinct orders) and overlay (instruments overlapping in a single arrangement).

Images were captured with a Logitech C920 webcam at 640×480 px, mounted 30 cm above the work surface on a fixed arm. Brightness was controlled by manually adjusting a desk lamp; lumen estimates were measured with a phone-based lux meter. Each image was annotated with bounding boxes using the TrayGuard webcam capture tool (`scripts/collect_data.py`) and exported as YOLO-format `.txt` files.

Seven staged train/test splits were constructed from the collected images (Table 37). All separated-condition stages trained on `order1` and `order2` layouts and tested on the same layouts; the overlay stage tested on the `overlay` layout.

Table 37: Brightness experiment stage definitions: brightness ranks, backgrounds, and layouts for each staged split

Stage	Train ranks	Test ranks	Train bg	Test bg
`brightest_train_darker_test`	1	2–11	both	both
`darkest_train_brighter_test`	11	1–10	both	both
`bright_train_dim_test`	1–6	7–11	both	both
`dim_train_bright_test`	7–11	1–6	both	both
`matte_train_reflective_test`	1–11	1–11	matte	reflective
`reflective_train_matte_test`	1–11	1–11	reflective	matte
`separated_train_overlay_test`	1–11	1–11	both	both

Rank 1 = 800 lumens; rank 11 = 80 lumens. "both" includes matte and reflective backgrounds.

Each stage was trained from COCO-pretrained YOLO11s at 640×640 px for 100 epochs with batch 16, SGD optimizer (initial lr 0.01, weight decay 0.0005), deterministic seed 42, and no data augmentation. Evaluation used confidence threshold 0.25 and IoU threshold 0.50 for class-aware greedy matching. No test-time augmentation was applied. Training took approximately 2 hours per stage on an NVIDIA RTX 3090.

Listing 1: Train all seven brightness stages

```
uv run python scripts/train_brightness_yolo.py --datasets-dir data/cv/brightness_experiment --device 0
```

Listing 2: Validate brightness stages and export per-image metrics

```
uv run python scripts/validate_brightness_experiments.py --datasets-dir data/cv/brightness_experiment --
    output-dir reports/brightness_validation
```

Figure 7 shows example images from the separated training set and the overlay test set, illustrating the layout gap that produced the largest performance degradation.

(a) Separated layout (training condition)

(b) Overlay layout (test condition)

Figure 7: Example images from the separated-to-overlay experiment. The training set uses non-overlapping layouts; the test set uses overlapping layouts.

A.2 Real-Instrument Experiments

The surgical kit proxy dataset contains six real instrument classes (scissor1, scissor2, scissor3, scissor4, forcep, scalpel) photographed on matte and reflective backgrounds across four sessions. Each image contains exactly six instruments (one per class) in fixed positions per session. Three staged experiments were trained with YOLO11s for 100 epochs.

Session-level splits were used for each experiment: `matte_train_reflective_test` trained on two matte-background sessions and tested on two reflective-background sessions; `reflective_train_matte_test` reversed the backgrounds; `shape_similarity_real_proxy` trained on two sessions and tested on a different held-out session. Training used the same configuration as the brightness experiments (YOLO11s, 640×640, 100 epochs, batch 16, deterministic seed 42, no augmentation).

Table 38 shows the per-class AP50 breakdown. The pattern is diagnostic of position-cue learning: classes that achieve high AP in one split fall to zero in others, matching session-specific instrument positions rather than visual features.

Table 38: Per-class AP50 for the three real-instrument experiments

Class	matte→reflective	reflective→matte	shape_similarity
scissor1	0.000	0.000	**0.862**
scissor2	0.435	0.000	0.000
scissor3	0.000	0.000	0.000
scissor4	0.000	0.000	0.587
forcep	0.517	0.000	0.000
scalpel	0.446	**0.717**	0.069

Figure 8 shows the most striking failure mode: the `reflective_train_matte_test` model detects only the scalpel (the most visually distinct instrument) and misses all five remaining classes. The confusion matrix for the `shape_similarity_real_proxy` experiment (Figure 9) confirms that class associations are split-specific rather than shape-driven.

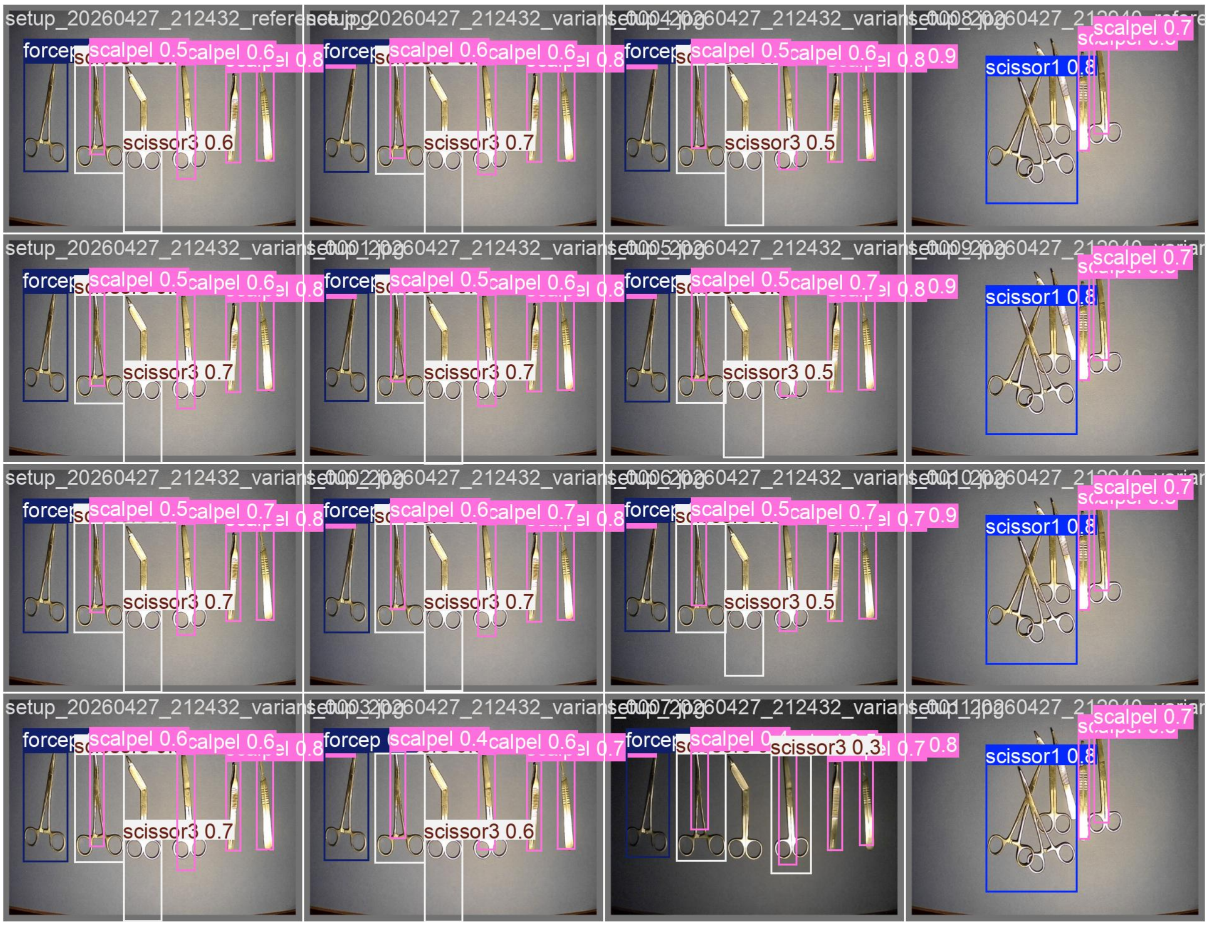


Figure 8: Prediction example from `reflective_train_matte_test`. Only the scalpel is detected (bounding box); the other five instruments (scissor1–4, forcep) are all missed.

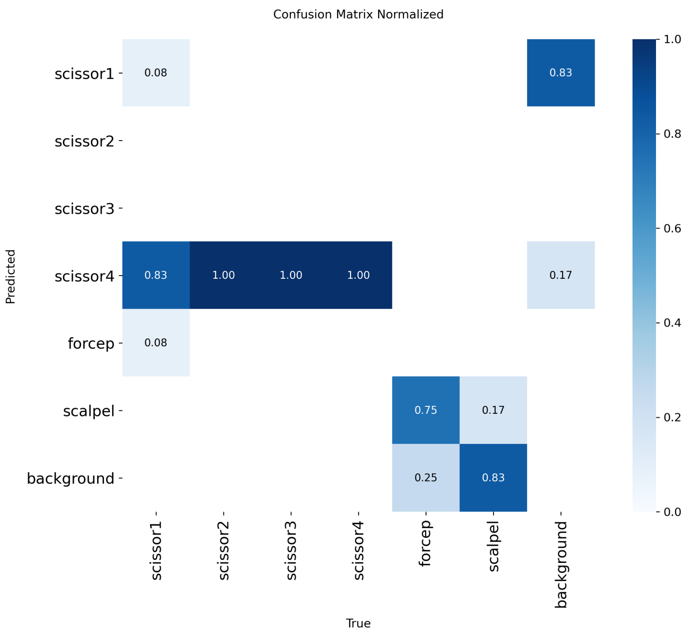


Figure 9: Normalized confusion matrix for `shape_similarity_real_proxy`. scissor1 and scissor4 are the only classes with nonzero true-positive rates; the remaining classes are uniformly missed.

Listing 3: Train the three surgical kit stages

```
uv run bash scripts/train_surgical_kit.sh
```

Listing 4: Re-evaluate all surgical kit stages

```
uv run python scripts/eval_surgical_kit.py
```

A.3 Lighting Diversity Experiment

Three YOLO11s models were trained on a self-collected spoon dataset (three classes: spoon1, spoon2, spoon3) with increasing lighting diversity: reference desk lamp only, reference plus 45-degree handheld flashlight, and reference plus both 45- and 90-degree flashlight. Each model was trained for 100 epochs at 640×640. Held-out test images were captured under intermediate flashlight angles not seen in any training set. Each additional lighting condition improved all metrics (Table 39).

Table 39: Lighting diversity experiment results

Training conditions	P	R	mAP50	mAP50-95
Reference only	0.951	0.998	0.995	0.923
Reference + 45	0.998	1.000	0.995	0.987
Reference + 45+ 90	0.999	1.000	0.995	0.990

Listing 5: Train lighting diversity models

```
uv run bash scripts/train_lighting_diversity.sh
```

Listing 6: Re-evaluate lighting diversity models

```
uv run python scripts/eval_lighting_diversity.py
```

Appendix B: Bills of Materials

Table 40: Bills of materials

Item	Description	Qty	Unit	Supplier	Unit Cost	Extended
1	Cardstock paper (letter, 65 lb)	1	pack (50)	Office supply	$8	$8
2	Clear adhesive sleeves	1	pack (100)	Office supply	$10	$10
3	Foam backing board (20×30 in)	1	sheet	Craft store	$5	$5
4	AprilTag markers (printed on paper)	30	sheets	In-house	$0	$0
5	Laptop or tablet for scanning	1	unit	Existing	$0	$0
6	Camera stand (optional)	1	unit	Online	$20	$20
7	Printed study flashcards	30	sheets	In-house	$0	$0
8	Assorted office supplies (tape, scissors, binder clips)	1	set	Office supply	$10	$10
Total prototype material cost						**$53**

Alternative material options and their current market prices are documented as cost-reference sources: stainless steel work tables [?, ?], stainless steel sheet stock [?], HDPE cutting board [?], camera copy stands [?, ?], surface liners [?, ?], board materials [?, ?, ?], and CPI-adjusted cost references [?]. These reflect upper-bound estimates; actual spending can be lower by reusing existing equipment or substituting available materials.

The primary project cost is labor: module authoring, software implementation, experimental data collection, study moderation, analysis, and report writing. These are captured in the project plan's task assignments and are not itemized as direct material purchases. The prototype is designed to work with materials already available in a typical academic or office environment.

Appendix C: Collaboration Task Matrix

Table 41: Collaboration task matrix

Section	Owen	Matthew	Andy	Jacob	Qiyu
ğ1. Executive Summary		✓			
ğ2. Introduction And Project Overview		✓	✓	✓	✓
ğ3. Problem Definition And Scope		✓	✓		✓
ğ4. First Attempt: A CV Tray Checker		✓	✓		
ğ5. Limitations of the CV Approach		✓	✓	✓	✓
ğ6. Literature And Related Work		✓	✓	✓	✓
ğ7. Requirements Definition		✓	✓	✓	✓
ğ8. Final Product Concept And Design Rationale		✓	✓		✓
ğ9. System Design		✓		✓	
ğ10. Analysis		✓			✓
ğ11. Evaluation And Validation Plan		✓			
ğ12. Risk Assessment And Scope Boundaries		✓			
ğ13. Broader Impacts And Ethics		✓		✓	✓
ğ14. Project Plan And Future Work		✓			
ğ15. Discussion And Final Recommendations		✓			
Appendices (A–C)		✓	✓	✓	✓
Support	✓				